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Preface

Linear Algebra is often the first place where a student encounters the true power of mathematical abstraction. In an introductory course, one learns to manipulate matrices, solve systems of linear equations, and perhaps calculate eigenvalues. But *Advanced* Linear Algebra is about something deeper: it is the study of structures.

The Philosophy of this Book

The central theme of these notes is the transition from concrete calculation to abstract reasoning. We begin not with numbers, but with axioms. By choosing to define mathematical objects—be they vectors, linear transformations, or inner products—entirely by how they behave rather than what they are, we unlock a universal language.

A theorem proved for an abstract vector space is a theorem proved simultaneously for geometric arrows, for polynomials, for continuous functions, and for quantum states. This “solve once, apply everywhere” philosophy is the heartbeat of modern mathematics.

How to Use These Notes

This book is designed to be a rigorous companion for a second course in linear algebra. Each chapter is structured to build a solid theoretical foundation:

- **Rigorous Definitions:** We start with precise mathematical statements to ensure no ambiguity in our logic.
- **Complete Proofs:** Every major result is accompanied by a full proof. Mathematics is not a spectator sport; I encourage you to attempt the proofs yourself before reading the solutions.
- **Illustrative Examples:** Abstract theory is balanced with concrete examples to build intuition.
- **Exercises:** Practice problems are provided at the end of sections to test your understanding.

Prerequisites

To get the most out of these notes, you should have a solid grasp of:

- **Introductory Linear Algebra:** You should be comfortable with matrix multiplication, determinants, and Gaussian elimination.

- **Calculus:** Familiarity with derivatives and integrals is assumed for some of our more advanced examples.
 - **Mathematical Maturity:** A basic understanding of logic and proof techniques (induction, contradiction, contrapositive) is essential. These are reviewed briefly in Chapter 0.
-

Welcome to the world of abstract linear algebra. I hope you find as much beauty in these structures as I do.

Chapter 0

0.1 Preliminary

This chapter covers the essential background needed for the study of linear algebra. Feel free to skip sections that you are already familiar with. This chapter serves only as a quick recap of the necessary mathematical background, so it might not be the best place to start if you are fresh to these topics. If you require a more detailed construction of these topics, I recommend checking out the first chapter of *Topology* by James Raymond Munkres (2nd ed.).

In addition to the following knowledge presented in this chapter, basic knowledge of logic and proof techniques is recommended, including:

- Direct proof
- Proof by contradiction
- Proof by induction
- Proof by contrapositive
- Proof by cases (exhaustion)
- Constructive proof (constructive existence)
- Non-constructive existence proof
- Uniqueness proof
- Proof of equivalence (iff proof)
- Disproof by counterexample
- Pigeonhole principle

0.2 Logic and Quantifiers

Mathematical statements in linear algebra often involve claims about “all” vectors or the “existence” of a solution. For instance, saying a matrix equation $\mathbf{Ax} = \mathbf{b}$ has a solution means *there exists* a vector $\mathbf{x}$ satisfying the equation; saying a set is linearly independent means *for all* nontrivial linear combinations, the result is nonzero. To work with these statements precisely—especially when proving or disproving them—we need the language of **quantifiers**.

0.2.1 Quantifiers

Definition 0.2.1 — Quantifiers

Let $P(x)$ be a statement (predicate) that depends on a variable x ranging over some set S .

- The **universal quantifier** $\forall$ means “for all” or “for every.” The statement

$$\forall x \in S : P(x)$$

asserts that $P(x)$ is true for *every* element x in S .

- The **existential quantifier** $\exists$ means “there exists” or “for some.” The statement

$$\exists x \in S : P(x)$$

asserts that $P(x)$ is true for *at least one* element x in S .

Remark. We sometimes write $\forall x \in S, P(x)$ or $(\forall x \in S) P(x)$ instead of $\forall x \in S : P(x)$. The notation $\exists! x \in S : P(x)$ means “there exists a **unique** x in S such that $P(x)$.”

Example. (Quantified Statements)

Consider the following statements:

1. $\forall x \in \mathbb{R} : x^2 \geq 0$.
This says “every real number squared is nonnegative.” This is **true**.
2. $\exists x \in \mathbb{R} : x^2 = -1$.
This says “there exists a real number whose square is -1 .” This is **false** in $\mathbb{R}$, but becomes true if we replace $\mathbb{R}$ with $\mathbb{C}$.
3. $\forall x \in \mathbb{R} : x < x + 1$.
This is **true** for all real numbers.
4. $\exists n \in \mathbb{Z} : n^2 = 2$.
This is **false**—there is no integer whose square is 2.

Quantifiers can be **nested**: statements may involve multiple quantifiers in sequence. The **order matters**.

Example. (Nested Quantifiers)

Consider these two statements about real numbers:

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : y > x$.
“For every real number x , there exists a real number y greater than x .” This is **true**—given any x , we can take $y = x + 1$.
2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R} : y > x$.
“There exists a real number y that is greater than all real numbers.” This is **false**—no such largest real number exists.

The order of quantifiers changes the meaning entirely.

0.2.2 Negating Quantified Statements

The key to proof by contradiction is knowing how to **negate** a statement. When negating quantified statements, there is a systematic rule: the universal quantifier becomes existential, and vice versa, while the predicate gets negated.

Definition 0.2.2 — Negation of Quantifiers

The negation of a quantified statement follows these rules:

$$\neg(\forall x \in S : P(x)) \equiv \exists x \in S : \neg P(x)$$

$$\neg(\exists x \in S : P(x)) \equiv \forall x \in S : \neg P(x)$$

In words:

- “Not all x satisfy P ” is equivalent to “there exists an x that does not satisfy P .”
- “There does not exist an x satisfying P ” is equivalent to “all x fail to satisfy P .”

Example. (Basic Negation Examples)

Here are two simple examples of negating quantified statements:

1. The negation of “ $\forall x \in \mathbb{R} : x^2 \geq 0$ ” is “ $\exists x \in \mathbb{R} : x^2 < 0$.”
The original is true; the negation is false.
2. The negation of “ $\exists n \in \mathbb{N} : n$ is even” is “ $\forall n \in \mathbb{N} : n$ is odd.”
The original is true (take $n = 2$); the negation is false.

The following examples demonstrate how to negate definitions that you will encounter repeatedly in linear algebra.

Example. (Negating the Definition of Injectivity)

Recall that a function $f : A \rightarrow B$ is **injective** if:

$$\forall a_1, a_2 \in A : f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$

To prove f is **not** injective, we negate this statement. Since $P \Rightarrow Q$ is equivalent to $\neg P \vee Q$, its negation is $P \wedge \neg Q$. Thus:

$$\exists a_1, a_2 \in A : f(a_1) = f(a_2) \wedge a_1 \neq a_2$$

To show a function is not injective, find two **distinct** elements that map to the same value.

Example. (Negating the Definition of Surjectivity)

A function $f : A \rightarrow B$ is **surjective** if:

$$\forall b \in B, \exists a \in A : f(a) = b$$

The negation is:

$$\exists b \in B, \forall a \in A : f(a) \neq b$$

To prove f is **not** surjective, find a single element in the codomain that no element of the domain maps to.

Remark. Mastering quantifier negation is essential for proof-based mathematics. Many proofs in linear algebra fall into one of the following patterns:

- Showing something holds for “all” vectors: use a direct proof starting with “let v be an arbitrary vector...”
- Showing something “exists”: construct it explicitly, or use contradiction by assuming nothing exists.
- Disproving a “for all” statement: find a single counterexample.
- Disproving an “exists” statement: show that every candidate fails.

0.3 Sets and Notation

Sets are the foundation of modern mathematics. In linear algebra, we study *vector spaces*, which are sets equipped with operations; *subspaces*, which are subsets with special properties; and *linear maps*, which are functions between sets. This section establishes the notation and operations we will use throughout.

0.3.1 Number Sets

We begin by defining the standard sets of numbers used throughout this book.

Definition 0.3.1 — Common Number Sets

We use the following notation for common sets of numbers:

- $\mathbb{N}$: The set of natural numbers $\{1, 2, 3, \dots\}$.
- $\mathbb{Z}$: The set of integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$.
- $\mathbb{Q}$: The set of rational numbers.
- $\mathbb{R}$: The set of real numbers.
- $\mathbb{C}$: The set of complex numbers.

Definition 0.3.2 — Cartesian Product

Let A and B be sets. The **Cartesian product** of A and B , denoted $A \times B$, is the set

of all ordered pairs:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

More generally, for sets $A_1, A_2, \dots, A_n$, the Cartesian product is:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Definition 0.3.3 — n -Tuple

An n -**tuple** is an ordered list of n elements. For a set A , the set of all n -tuples with entries from A is denoted:

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}} = \{(a_1, a_2, \dots, a_n) : a_i \in A \text{ for } i = 1, 2, \dots, n\}$$

Example. (n -Tuples)

Here are examples of n -tuples from various sets:

1. $\mathbb{R}^2$: The Cartesian plane. Elements include $(0, 0)$, $(1, 2)$, $(-3, \pi)$, $(\sqrt{2}, -5.7)$.
2. $\mathbb{R}^3$: 3-dimensional space. Elements include $(1, 0, 0)$, $(1, 2, 3)$, $(-1, \pi, e)$.
3. $\mathbb{Z}^2$: Pairs of integers. Elements include $(0, 0)$, $(1, -2)$, $(5, 7)$. Note that $(\frac{1}{2}, 3) \notin \mathbb{Z}^2$.
4. $\mathbb{C}^2$: Pairs of complex numbers. Elements include $(1 + i, 2 - 3i)$, $(i, 0)$, $(3, 4)$.
5. $\{0, 1\}^3$: Binary 3-tuples. This set has exactly 8 elements:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

In general, if A is a finite set with $|A| = k$ elements, then A^n has k^n elements.

0.3.2 Set-Builder Notation and the Empty Set

When defining sets, we often describe them by specifying a property that their elements must satisfy.

Definition 0.3.4 — Set-Builder Notation

Let S be a set and $P(x)$ a predicate. The **set-builder notation**

$$\{x \in S : P(x)\}$$

denotes the set of all elements x in S for which $P(x)$ is true. Variants include $\{x \in S \mid P(x)\}$ and $\{x \in S; P(x)\}$.

Example. (Set-Builder Examples)

The following sets are defined using set-builder notation:

1. $\{x \in \mathbb{R} : x^2 < 4\} = (-2, 2)$, the open interval from -2 to 2 .
2. $\{n \in \mathbb{Z} : n \text{ is even}\} = \{\dots, -4, -2, 0, 2, 4, \dots\}$.
3. $\{(x, y) \in \mathbb{R}^2 : x + y = 1\}$ is the line through $(1, 0)$ and $(0, 1)$.

Definition 0.3.5 — Empty Set

The **empty set**, denoted $\emptyset$ or $\{\}$, is the set containing no elements:

$$\emptyset = \{x : x \neq x\}$$

For any element a , we have $a \notin \emptyset$.

Remark. Statements of the form “ $\forall x \in \emptyset : P(x)$ ” are **vacuously true**—there are no elements to check. This technicality appears when discussing trivial subspaces or the span of an empty collection.

0.3.3 Subsets

The concept of a *subset* is central to linear algebra. When we define a subspace, we first verify it is a subset of the ambient space.

Definition 0.3.6 — Subset

Let A and B be sets. We say A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B :

$$A \subseteq B \iff \forall x : (x \in A \Rightarrow x \in B)$$

If $A \subseteq B$ and $A \neq B$, we say A is a **proper subset** of B , written $A \subsetneq B$ or $A \subset B$.

Example. (Subset Examples)

The following illustrate subset relationships:

1. $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$.
2. $\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2\} \subsetneq \{1, 2, 3\}$.
3. For any set A : $\emptyset \subseteq A$ and $A \subseteq A$.
4. $\{x \in \mathbb{R} : x^2 = 1\} = \{-1, 1\} \subseteq \mathbb{Z}$.

Definition 0.3.7 — Set Equality

Two sets A and B are **equal**, written $A = B$, if they contain exactly the same elements:

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A)$$

Remark. To prove two sets are equal, we typically prove **two subset inclusions**: first show $A \subseteq B$, then show $B \subseteq A$. This technique appears constantly when proving that two subspaces or spans are equal.

0.3.4 Set Operations

Sets can be combined using union, intersection, and difference. These operations appear when discussing sums of subspaces and their intersections.

Definition 0.3.8 — Union

The **union** of sets A and B is the set of elements in A or B (or both):

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Definition 0.3.9 — Intersection

The **intersection** of sets A and B is the set of elements in both A and B :

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Two sets are **disjoint** if $A \cap B = \emptyset$.

Definition 0.3.10 — Set Difference

The **set difference** (or **relative complement**) of B in A is:

$$A \setminus B = \{x \in A : x \notin B\}$$

When $B \subseteq U$ for some universal set U , the **complement** of B in U is $B^c = U \setminus B$.

Example. (Set Operations)

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then:

- $A \cup B = \{1, 2, 3, 4, 5, 6\}$
- $A \cap B = \{3, 4\}$
- $A \setminus B = \{1, 2\}$
- $B \setminus A = \{5, 6\}$

Example. (Set Operations with Intervals)

Let $A = [0, 2]$ and $B = [1, 3]$ be intervals in $\mathbb{R}$. Then:

- $A \cup B = [0, 3]$
- $A \cap B = [1, 2]$
- $A \setminus B = [0, 1)$

Remark. In linear algebra, we will see that the **intersection** of two subspaces is always a subspace, but the **union** of two subspaces is generally *not* a subspace (this is a common exam question!). Instead of union, we use the **sum** of subspaces: $U + W = \{u + w : u \in U, w \in W\}$.

0.4 Functions

Functions are the mathematical way to describe *processes* or *transformations*—rules that take an input and produce an output. In linear algebra, the central objects of study are **linear maps** (also called linear transformations), which are functions between vector spaces that respect the vector space structure. Understanding functions in general—especially their composition and inverses—is crucial because:

- **Matrix multiplication** corresponds to **composition** of linear maps.
- An **invertible matrix** corresponds to a **bijective** linear map.
- Many proofs involve showing a linear map is injective, surjective, or bijective.

0.4.1 Definition and Basic Properties

Definition 0.4.1 — Function

A **function** f from a set A to a set B , denoted $f : A \rightarrow B$, is a rule that assigns to each element $a \in A$ exactly one element $f(a) \in B$. The set A is called the **domain** and B is called the **codomain**.

Definition 0.4.2 — Injective, Surjective, Bijective

Let $f : A \rightarrow B$ be a function.

- f is **injective** (one-to-one) if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.
- f is **surjective** (onto) if for every $b \in B$, there exists $a \in A$ such that $f(a) = b$.
- f is **bijective** if it is both injective and surjective

Example. (Injective, Surjective, and Bijective Functions)

Consider the following functions:

1. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$ is **bijective**.
 - Injective: If $2x_1 + 1 = 2x_2 + 1$, then $x_1 = x_2$.
 - Surjective: For any $y \in \mathbb{R}$, we have $f\left(\frac{y-1}{2}\right) = y$.

2. $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$ is **neither injective nor surjective**.
 - Not injective: $g(1) = g(-1) = 1$, but $1 \neq -1$.
 - Not surjective: There is no $x \in \mathbb{R}$ such that $g(x) = -1$.
3. $h : \mathbb{R} \rightarrow [0, \infty)$, $h(x) = x^2$ is **surjective but not injective**.
 - Not injective: $h(1) = h(-1) = 1$.
 - Surjective: For any $y \geq 0$, we have $h(\sqrt{y}) = y$.
4. $k : [0, \infty) \rightarrow \mathbb{R}$, $k(x) = x^2$ is **injective but not surjective**.
 - Injective: If $x_1^2 = x_2^2$ with $x_1, x_2 \geq 0$, then $x_1 = x_2$.
 - Not surjective: There is no $x \geq 0$ such that $k(x) = -1$.

0.4.2 Function Composition

When we apply one function after another, we obtain a **composition**. This concept is fundamental: matrix multiplication is precisely the composition of the corresponding linear maps.

Definition 0.4.3 — Function Composition

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. The **composition** of g with f , denoted $g \circ f$, is the function $g \circ f : A \rightarrow C$ defined by:

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A$$

We read $g \circ f$ as “ g composed with f ” or “ g after f .”

Remark. The notation $g \circ f$ means “first apply f , then apply g .” The order matters and can be confusing—the function written on the right is applied first. This mirrors how matrix multiplication works: in $\mathbf{ABx}$, we first multiply $\mathbf{Bx}$, then multiply by $\mathbf{A}$.

Example. (Function Composition)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = x + 1$. We compute both compositions:

- $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1$
- $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2$

Note that $g \circ f \neq f \circ g$ in general—composition is **not commutative**.

Theorem 0.4.1 — Associativity of Composition

Let $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$ be functions. Then:

$$(h \circ g) \circ f = h \circ (g \circ f)$$

That is, function composition is **associative**.

Proof. For any $a \in A$:

$$((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a)))$$

and

$$(h \circ (g \circ f))(a) = h((g \circ f)(a)) = h(g(f(a)))$$

Since both expressions equal $h(g(f(a)))$ for all $a \in A$, the compositions are equal. ■

Theorem 0.4.2 — Composition Preserves Injectivity and Surjectivity

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

1. If f and g are both injective, then $g \circ f$ is injective.
2. If f and g are both surjective, then $g \circ f$ is surjective.
3. If f and g are both bijective, then $g \circ f$ is bijective.

Proof. We prove (1). Suppose f and g are injective and $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. Thus $g \circ f$ is injective.

The proof of (2) and (3) are similar exercises. ■

0.4.3 Identity Function

Definition 0.4.4 — Identity Function

For any set A , the **identity function** $\text{id}_A : A \rightarrow A$ is defined by:

$$\text{id}_A(a) = a \quad \text{for all } a \in A$$

Remark. The identity function satisfies $f \circ \text{id}_A = f$ for any $f : A \rightarrow B$, and $\text{id}_B \circ f = f$. It is the “do nothing” function. In matrix terms, the identity matrix $\mathbf{I}_n$ represents the identity function on F^n .

0.4.4 Inverse Functions

A bijective function can be “undone”—this leads to the concept of an inverse function. Understanding inverses is essential because **invertible matrices** are precisely those representing bijective linear maps.

Definition 0.4.5 — Inverse Function

Let $f : A \rightarrow B$ be a function. A function $g : B \rightarrow A$ is called an **inverse** of f if:

$$g \circ f = \text{id}_A \quad \text{and} \quad f \circ g = \text{id}_B$$

If such a g exists, we say f is **invertible** and write $g = f^{-1}$.

Theorem 0.4.3 — Bijective if and only if Invertible

A function $f : A \rightarrow B$ is bijective if and only if it has an inverse $f^{-1} : B \rightarrow A$.

Moreover, if the inverse exists, it is unique.

Proof. ($\Rightarrow$) Suppose f is bijective. For each $b \in B$, since f is surjective, there exists some $a \in A$ with $f(a) = b$. Since f is injective, this a is unique. Define $g(b) = a$. We verify that g is an inverse:

- $(g \circ f)(a) = g(f(a)) = a$, so $g \circ f = \text{id}_A$.
- $(f \circ g)(b) = f(g(b)) = f(a) = b$, so $f \circ g = \text{id}_B$.

Thus $g = f^{-1}$ is an inverse of f .

($\Leftarrow$) Suppose f has an inverse g . We show f is bijective:

- *Injective:* If $f(a_1) = f(a_2)$, then $a_1 = g(f(a_1)) = g(f(a_2)) = a_2$.
- *Surjective:* For any $b \in B$, we have $f(g(b)) = b$, so b is in the image of f .

Uniqueness: If g and h are both inverses of f , then:

$$g = g \circ \text{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_A \circ h = h$$

■

Example. (Inverse Functions)

The following examples illustrate inverse functions and when they do or do not exist:

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + 3$. This is bijective, and its inverse is $f^{-1}(y) = \frac{y-3}{2}$.
Verification: $f(f^{-1}(y)) = 2 \cdot \frac{y-3}{2} + 3 = y$ and $f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = x$.
2. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = x^2$. This is not bijective (not injective), so g^{-1} does not exist as a function $\mathbb{R} \rightarrow \mathbb{R}$.
3. Let $h : [0, \infty) \rightarrow [0, \infty)$ be defined by $h(x) = x^2$. With restricted domain and codomain, h is bijective, and $h^{-1}(y) = \sqrt{y}$.

Theorem 0.4.4 — Inverse of a Composition

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are both bijective, then $g \circ f$ is bijective and:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Proof. We verify the inverse property:

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_B \circ f = f^{-1} \circ f = \text{id}_A$$

Similarly, $(g \circ f) \circ (f^{-1} \circ g^{-1}) = \text{id}_C$. ■

Remark. Compare this to the matrix identity $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ for invertible matrices. The reversal of order is the same phenomenon: to undo “first f , then g ,” you must “first undo g , then undo f .”

0.5 Equivalence Relations

In mathematics, we often want to say that two objects are “the same” in some sense, even if they are not identical. For instance:

- Two fractions $\frac{1}{2}$ and $\frac{2}{4}$ represent the same rational number.
- Two matrices $\mathbf{A}$ and $\mathbf{B}$ are **similar** if $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ for some invertible $\mathbf{P}$ —they represent the same linear transformation in different bases.
- Two vector spaces are **isomorphic** if there exists a bijective linear map between them—they have the same “structure.”

Each of these notions of “sameness” is captured by an **equivalence relation**. Understanding equivalence relations helps us recognize when different-looking objects are fundamentally alike.

0.5.1 Relations

Definition 0.5.1 — Relation

A **(binary) relation** on a set A is a subset $R \subseteq A \times A$. If $(a, b) \in R$, we write $a \sim b$ (or $a R b$) and say “ a is related to b .”

Example. (Examples of Relations)

Here are some common examples of relations:

1. On $\mathbb{Z}$, define $a \sim b$ if $a \leq b$. This is the “less than or equal to” relation.
2. On $\mathbb{R}$, define $a \sim b$ if $|a - b| < 1$. This says a and b are “close.”

3. On the set of all people, define $a \sim b$ if a and b have the same birthday.

0.5.2 Equivalence Relations

Not all relations behave like “sameness.” An **equivalence relation** is one that satisfies three natural properties that any notion of sameness should have.

Definition 0.5.2 — Equivalence Relation

A relation $\sim$ on a set A is an **equivalence relation** if it satisfies:

1. **Reflexivity:** $\forall a \in A : a \sim a$.
(Every element is related to itself.)
2. **Symmetry:** $\forall a, b \in A : a \sim b \Rightarrow b \sim a$.
(If a is related to b , then b is related to a .)
3. **Transitivity:** $\forall a, b, c \in A : (a \sim b \wedge b \sim c) \Rightarrow a \sim c$.
(If a is related to b and b is related to c , then a is related to c .)

Example. (Examples of Equivalence Relations)

The following are all equivalence relations:

1. **Equality:** On any set A , the relation $a \sim b \iff a = b$ is an equivalence relation (the finest one possible).
2. **Congruence modulo n :** On $\mathbb{Z}$, define $a \equiv b \pmod{n}$ if $n \mid (a - b)$.
 - Reflexive: $n \mid 0$, so $a \equiv a$.
 - Symmetric: If $n \mid (a - b)$, then $n \mid (b - a)$.
 - Transitive: If $n \mid (a - b)$ and $n \mid (b - c)$, then $n \mid ((a - b) + (b - c)) = (a - c)$.
3. **Same birthday:** On the set of people, “ a has the same birthday as b ” is an equivalence relation.
4. **Same absolute value:** On $\mathbb{R}$, define $a \sim b \iff |a| = |b|$. This is an equivalence relation.

Example. (Non-Examples)

The following relations fail to be equivalence relations:

1. On $\mathbb{Z}$, the relation $a \leq b$ is **not** an equivalence relation:
 - Reflexive: Yes, $a \leq a$.
 - Symmetric: **No.** $1 \leq 2$ but $2 \not\leq 1$.
2. On $\mathbb{R}$, the relation $|a - b| < 1$ is **not** an equivalence relation:
 - Reflexive: Yes, $|a - a| = 0 < 1$.
 - Symmetric: Yes.
 - Transitive: **No.** We have $|0 - 0.6| < 1$ and $|0.6 - 1.2| < 1$, but $|0 - 1.2| = 1.2 \not< 1$.

0.5.3 Equivalence Classes

An equivalence relation partitions a set into disjoint groups of “equivalent” elements. Each such group is called an **equivalence class**.

Definition 0.5.3 — Equivalence Class

Let $\sim$ be an equivalence relation on A . The **equivalence class** of an element $a \in A$ is the set of all elements equivalent to a :

$$[a] = \{b \in A : b \sim a\}$$

The element a is called a **representative** of the class $[a]$.

Example. (Examples of Equivalence Classes)

We illustrate equivalence classes with two examples:

1. **Congruence modulo 3:** The equivalence classes of $\mathbb{Z}$ under $\equiv \pmod{3}$ are:

$$[0] = \{\dots, -6, -3, 0, 3, 6, 9, \dots\}$$

$$[1] = \{\dots, -5, -2, 1, 4, 7, 10, \dots\}$$

$$[2] = \{\dots, -4, -1, 2, 5, 8, 11, \dots\}$$

Note that $[0] = [3] = [6] = \dots$ and $[1] = [4] = [7] = \dots$. Different representatives can name the same class.

2. **Same absolute value on $\mathbb{R}$:** The equivalence class of 2 is $[2] = \{-2, 2\}$. The equivalence class of 0 is $[0] = \{0\}$.

Theorem 0.5.1 — Equivalence Classes Partition the Set

Let $\sim$ be an equivalence relation on A . Then:

1. Every element belongs to some equivalence class: $a \in [a]$.
2. Two equivalence classes are either equal or disjoint: $[a] = [b]$ or $[a] \cap [b] = \emptyset$.
3. The union of all equivalence classes is A .

In other words, the equivalence classes form a **partition** of A .

Proof.

- (1) By reflexivity, $a \sim a$, so $a \in [a]$.
- (2) Suppose $[a] \cap [b] \neq \emptyset$, and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$. By transitivity, $a \sim b$.

Now we show $[a] = [b]$. Let $x \in [a]$, so $x \sim a$. Since $a \sim b$, by transitivity $x \sim b$, so $x \in [b]$. Thus $[a] \subseteq [b]$. By symmetry, $[b] \subseteq [a]$, so $[a] = [b]$.

(3) Follows from (1): every element a is in $[a]$.

■

Definition 0.5.4 — Quotient Set

Let $\sim$ be an equivalence relation on A . The **quotient set** (or **quotient of A by $\sim$**) is the set of all equivalence classes:

$$A/\sim = \{[a] : a \in A\}$$

Example. (Quotient Sets)

The following are examples of quotient sets:

1. $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$ has three elements. This is often written $\mathbb{Z}/3\mathbb{Z}$ or $\mathbb{Z}_3$.
2. The quotient of $\mathbb{R}$ by “same absolute value” is in bijection with $[0, \infty)$ —each equivalence class is determined by its absolute value.

0.5.4 Equivalence Relations in Linear Algebra

The following equivalence relations will appear throughout your study of linear algebra:

Example. (Matrix Similarity)

On $M_n(F)$, define $\mathbf{A} \sim \mathbf{B}$ if there exists an invertible matrix $\mathbf{P} \in M_n(F)$ such that $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. This is called **similarity**.

- Reflexive: $\mathbf{A} = \mathbf{I}^{-1}\mathbf{A}\mathbf{I}$, so $\mathbf{A} \sim \mathbf{A}$.
- Symmetric: If $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, then $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1} = (\mathbf{P}^{-1})^{-1}\mathbf{B}(\mathbf{P}^{-1})$, so $\mathbf{B} \sim \mathbf{A}$.
- Transitive: If $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ and $\mathbf{C} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}$, then $\mathbf{C} = \mathbf{Q}^{-1}\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{Q} = (\mathbf{PQ})^{-1}\mathbf{A}(\mathbf{PQ})$.

Similar matrices represent the same linear transformation in different bases. They share many properties: determinant, trace, eigenvalues, and characteristic polynomial.

Example. (Row Equivalence)

Two matrices are **row equivalent** if one can be obtained from the other by a sequence of elementary row operations. This is an equivalence relation on $M_{m \times n}(F)$.

Remark. When you encounter a new equivalence relation, ask: “What properties do equivalent objects share?” For similarity, the answer includes eigenvalues and determinant. These shared properties are called **invariants** of the equivalence relation.

0.6 Polynomials

Definition 0.6.1 — Polynomial

A **polynomial** over a field F in the indeterminate x is an expression of the form:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{k=0}^n a_k x^k$$

where $a_0, a_1, \dots, a_n \in F$ are called the **coefficients** of $p(x)$.

Definition 0.6.2 — Degree

Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a nonzero polynomial with $a_n \neq 0$. The **degree** of $p(x)$, denoted $\deg p(x)$, is n . The coefficient a_n is called the **leading coefficient**.

By convention, the zero polynomial $p(x) = 0$ has degree $-\infty$.

Definition 0.6.3 — Polynomial Ring

The set of all polynomials over a field F is denoted $F[x]$:

$$F[x] = \left\{ \sum_{k=0}^n a_k x^k : n \in \mathbb{N}, a_k \in F \right\}$$

This set forms a **ring** under polynomial addition and multiplication. (You can omit what ring means here.)

Example. (Elements of the Polynomial Ring)

The following are elements of $\mathbb{R}[x]$:

$$3x^2 - 2x + 1, \quad x^5 + \pi x^3 - \sqrt{2}, \quad 7, \quad 0$$

Note that constant polynomials (including 0) are also polynomials.

In linear algebra, we often work with polynomials of bounded degree.

Definition 0.6.4 — Polynomials of Degree at Most n

Let F be a field and $n \in \mathbb{N}$. The set of all polynomials over F of degree at most n is denoted:

$$F[x]_{\leq n} = \{p(x) \in F[x] : \deg p(x) \leq n\}$$

Equivalently:

$$F[x]_{\leq n} = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 : a_k \in F\}$$

Example. (Polynomials of Degree at Most 2)

The set $\mathbb{R}[x]_{\leq 2}$ consists of all polynomials of degree at most 2:

$$\mathbb{R}[x]_{\leq 2} = \{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$$

Examples include $3x^2 - 2x + 1$, $5x + 7$, and -4 .

Definition 0.6.5 — Polynomials of Degree Exactly n

We use $F[x]_{=n}$ to denote polynomials of degree **exactly** n :

$$F[x]_{=n} = \{p(x) \in F[x] : \deg p(x) = n\}$$

Equivalently:

$$F[x]_{=n} = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 : a_n \neq 0, a_k \in F\}$$

Remark. Note the distinction:

- $F[x]_{\leq n}$ contains polynomials of degree $\leq n$ (including the zero polynomial)
- $F[x]_{=n}$ contains polynomials of degree **exactly** n (so $a_n \neq 0$)

Thus $F[x]_{\leq n} \supsetneq F[x]_{=n}$, and in fact $F[x]_{\leq n} = F[x]_{=n} \cup F[x]_{\leq n-1}$.

Example. (Comparing Notations)

The following examples highlight the difference between the notations introduced above:

- $\mathbb{R}[x]_{\leq 2} = \{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$ includes $x^2 + 1$, $3x - 2$, and 5
- $\mathbb{R}[x]_{=2} = \{ax^2 + bx + c : a \neq 0, a, b, c \in \mathbb{R}\}$ includes $x^2 + 1$ but **not** $3x - 2$ or 5

0.7 Fields

Definition 0.7.1 — Field

A set F is called a **field** if we can define two binary operations addition $+$, and multiplication $\cdot$

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F$$

such that:

1. $a + b = b + a$

2. $(a + b) + c = a + (b + c)$
3. There exists $0 \in F$ such that $a + 0 = a$.
4. For every $a \in F$, there exists $-a \in F$ such that $a + (-a) = 0$.
5. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
6. $a \cdot b = b \cdot a$
7. There exists $1 \in F$ such that $1 \neq 0$ and $a \cdot 1 = a$.
8. For every $a \in F \setminus \{0\}$, there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.
9. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Remark. We can shorten “ F is a field with operations $+$ and $\cdot$ ” to “ $(F, +, \cdot)$ is a field”. Moreover when the context of $+$ and $\cdot$ is clear, we simply say “ F is a field”, which we will do most of the time in this book.

Example. (Examples of Fields)

The following sets are examples (and non-examples) of fields:

1. $\mathbb{R}$ is a field.
2. $\mathbb{C}$ is a field.
3. $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$ is a field.
4. $\mathbb{Z}$ is **not** a field since **(F8)** fails.
5. $M_n(\mathbb{R})$ is **not** a field since **(F6)** fails.

Example. (Finite Fields)

Let $\mathbb{F}_n$ be the finite field with n elements. Then $(\mathbb{F}_2, +, \cdot)$ is defined to be:

$$\mathbb{F}_2 = \{0, 1\}, \quad \begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}, \quad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}.$$

Then one can verify that $(\mathbb{F}_2, +, \cdot)$ is a field.

0.8 Matrices

Definition 0.8.1 — Matrix

An $m \times n$ **matrix** over a field F is a rectangular array of elements from F arranged in m rows and n columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Remark. $M_{m \times n}(F)$ is the set of all matrices of size $m \times n$ and have entries in F . In the above definition, we can write $\mathbf{A} \in M_{m \times n}(F)$.

Moreover, when $m = n$, we abbreviate $M_{m \times n}(F)$ to $M_n(F)$.

Example. (Matrices)

Here are some examples of matrices:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in M_2(\mathbb{R}), \quad \begin{bmatrix} 1 & 0 & -1 \\ 2 & 5 & 3 \end{bmatrix} \in M_{2 \times 3}(\mathbb{R}), \quad \begin{bmatrix} i & 1+i \\ 0 & 2-i \end{bmatrix} \in M_2(\mathbb{C}).$$

0.8.1 Special Matrices

Definition 0.8.2 — Zero Matrix

The **zero matrix** $\mathbf{O} \in M_{m \times n}(F)$ is the matrix with all entries equal to zero:

$$\mathbf{O} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Definition 0.8.3 — Identity Matrix

The **identity matrix** $\mathbf{I}_n \in M_n(F)$ is the $n \times n$ square matrix with 1s on the diagonal and 0s elsewhere:

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

When the size is clear from context, we simply write $\mathbf{I}$.

Definition 0.8.4 — Transpose

Let $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$. The **transpose** of $\mathbf{A}$, denoted $\mathbf{A}^\top$, is the $n \times m$ matrix obtained by swapping rows and columns:

$$(\mathbf{A}^\top)_{ij} = a_{ji}.$$

Visually, the rows of $\mathbf{A}$ become the columns of $\mathbf{A}^\top$:

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}^\top = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}.$$

Definition 0.8.5 — Symmetric Matrix

A square matrix $\mathbf{A} \in M_n(F)$ is **symmetric** if $\mathbf{A} = \mathbf{A}^\top$. This means it is symmetric across its main diagonal:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}.$$

Definition 0.8.6 — Diagonal Matrix

A square matrix $\mathbf{A} \in M_n(F)$ is **diagonal** if all entries off the main diagonal are zero:

$$\mathbf{A} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}.$$

We often write $\mathbf{A} = \text{diag}(d_1, d_2, \dots, d_n)$.

Definition 0.8.7 — Upper Triangular Matrix

A square matrix $\mathbf{A} \in M_n(F)$ is **upper triangular** if all entries below the main diagonal are zero:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}.$$

Example. (Zero and Identity Matrices)

$$\mathbf{O}_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Example. (Examples of Special Matrices)

Here are concrete examples illustrating the special types of matrices defined above:

- **Transpose:** $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$

- **Symmetric:** $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ is symmetric since $a_{ij} = a_{ji}$.
- **Diagonal:** $\text{diag}(3, -1, 0) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
- **Upper Triangular:** $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.

0.8.2 Matrix Addition

Definition 0.8.8 — Matrix Addition

Let $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ be matrices in $M_{m \times n}(F)$. Their **sum** $\mathbf{A} + \mathbf{B}$ is the matrix $\mathbf{C} = (c_{ij}) \in M_{m \times n}(F)$ where:

$$c_{ij} = a_{ij} + b_{ij}.$$

Definition 0.8.9 — Additive Inverse

The **additive inverse** (or **negation**) of a matrix $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ is the matrix $-\mathbf{A} = (-a_{ij}) \in M_{m \times n}(F)$.

Example. (Matrix Addition)

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Then:

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

Theorem 0.8.1 — Properties of Matrix Addition

Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in M_{m \times n}(F)$. Then:

1. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ (Commutativity)
2. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$ (Associativity)
3. $\mathbf{A} + \mathbf{O} = \mathbf{A}$ (Additive identity)
4. $\mathbf{A} + (-\mathbf{A}) = \mathbf{O}$ (Additive inverse)

Proof. We prove commutativity. Let $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$. Then:

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (\mathbf{B} + \mathbf{A})_{ij}$$

where the second equality uses commutativity of addition in the underlying field. Since this holds for all i, j , we have $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.

The remaining properties follow similarly. ■

0.8.3 Matrix Multiplication

Definition 0.8.10 — Matrix Multiplication

Let $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ and $\mathbf{B} = (b_{jk}) \in M_{n \times p}(F)$. Their **product** $\mathbf{AB}$ is the matrix $\mathbf{C} = (c_{ik}) \in M_{m \times p}(F)$ where:

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$$

In other words, the (i, k) -entry of $\mathbf{AB}$ is the dot product of the i -th row of $\mathbf{A}$ with the k -th column of $\mathbf{B}$.

Example. (Matrix Multiplication)

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in M_2(\mathbb{R})$ and $\mathbf{B} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \in M_2(\mathbb{R})$. We compute $\mathbf{AB}$ entry by entry:

$$\begin{aligned} (\mathbf{AB})_{11} &= (1)(5) + (2)(7) = 5 + 14 = 19 \\ (\mathbf{AB})_{12} &= (1)(6) + (2)(8) = 6 + 16 = 22 \\ (\mathbf{AB})_{21} &= (3)(5) + (4)(7) = 15 + 28 = 43 \\ (\mathbf{AB})_{22} &= (3)(6) + (4)(8) = 18 + 32 = 50 \end{aligned}$$

Therefore:

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Example. (Matrix Multiplication with Different Dimensions)

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in M_{2 \times 3}(\mathbb{R})$ and $\mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \in M_{3 \times 1}(\mathbb{R})$. The product $\mathbf{AB} \in M_{2 \times 1}(\mathbb{R})$:

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} (1)(1) + (2)(0) + (3)(-1) \\ (4)(1) + (5)(0) + (6)(-1) \end{bmatrix} = \begin{bmatrix} 1 + 0 - 3 \\ 4 + 0 - 6 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

Note that $\mathbf{BA}$ is **not defined** since the number of columns of $\mathbf{B}$ (which is 1) does not equal the number of rows of $\mathbf{A}$ (which is 2).

Theorem 0.8.2 — Properties of Matrix Multiplication

Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be matrices of compatible dimensions. Then:

1. $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ (Associativity)
2. $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ (Left distributivity)
3. $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$ (Right distributivity)
4. $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$ (Multiplicative identity)

Proof. We prove associativity. Let $\mathbf{A}$ be $m \times n$, $\mathbf{B}$ be $n \times p$, and $\mathbf{C}$ be $p \times q$. For any entry (i, ℓ) :

$$\begin{aligned}
 ((\mathbf{AB})\mathbf{C})_{i\ell} &= \sum_{k=1}^p (\mathbf{AB})_{ik} c_{k\ell} = \sum_{k=1}^p \left(\sum_{j=1}^n a_{ij} b_{jk} \right) c_{k\ell} \\
 &= \sum_{k=1}^p \sum_{j=1}^n a_{ij} b_{jk} c_{k\ell} = \sum_{j=1}^n \sum_{k=1}^p a_{ij} b_{jk} c_{k\ell} \\
 &= \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^p b_{jk} c_{k\ell} \right) = \sum_{j=1}^n a_{ij} (\mathbf{BC})_{j\ell} \\
 &= (\mathbf{A}(\mathbf{BC}))_{i\ell}
 \end{aligned}$$

■

Remark. Matrix multiplication is **not** commutative in general. That is, $\mathbf{AB} \neq \mathbf{BA}$ even when both products are defined.

0.8.4 Scalar Multiplication

Definition 0.8.11 — Scalar Multiplication

Let $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ and $c \in F$ be a scalar. The **scalar multiple** $c\mathbf{A}$ is the matrix in $M_{m \times n}(F)$ with entries (ca_{ij}) .

0.8.5 Trace of a Matrix

Definition 0.8.12 — Trace

Let $\mathbf{A} = (a_{ij}) \in M_n(F)$ be a square matrix. The **trace** of $\mathbf{A}$, denoted $\text{tr}(\mathbf{A})$, is the sum of the entries on its main diagonal:

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Example. (Trace)

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Then $\text{tr}(\mathbf{A}) = 1 + 5 + 9 = 15$.

0.8.6 Properties of Transpose and Trace**Theorem 0.8.3 — Properties of Transpose**

Let $\mathbf{A}, \mathbf{B}$ be matrices of compatible sizes and $c \in F$. Then:

1. $(\mathbf{A}^\top)^\top = \mathbf{A}$
2. $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$
3. $(c\mathbf{A})^\top = c\mathbf{A}^\top$
4. $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ (Reversal rule)

Proof. We prove the reversal rule $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$. Let $\mathbf{A} \in M_{m \times n}(F)$ and $\mathbf{B} \in M_{n \times p}(F)$.

$$((\mathbf{AB})^\top)_{ij} = (\mathbf{AB})_{ji} = \sum_{k=1}^n a_{jk} b_{ki}$$

Now consider $\mathbf{B}^\top \mathbf{A}^\top$. Its (i, j) -entry is:

$$(\mathbf{B}^\top \mathbf{A}^\top)_{ij} = \sum_{k=1}^n (\mathbf{B}^\top)_{ik} (\mathbf{A}^\top)_{kj} = \sum_{k=1}^n b_{ki} a_{jk} = \sum_{k=1}^n a_{jk} b_{ki}$$

Since the entries are equal for all i, j , we have $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$. ■

Theorem 0.8.4 — Properties of Trace

Let $\mathbf{A}, \mathbf{B} \in M_n(F)$ and $c \in F$. Then:

1. $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$
2. $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$
3. $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A})$
4. $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ (Cyclic property)

Proof. We prove $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

$$\text{tr}(\mathbf{AB}) = \sum_{i=1}^n (\mathbf{AB})_{ii} = \sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} b_{ji} \right)$$

Rearranging the summation:

$$\mathrm{tr}(\mathbf{AB}) = \sum_{j=1}^n \sum_{i=1}^n b_{ji} a_{ij} = \sum_{j=1}^n (\mathbf{BA})_{jj} = \mathrm{tr}(\mathbf{BA}).$$

■

Chapter 1

1.1 Vector Spaces and Dimensions

In this chapter, we introduce the fundamental concept of a vector space and discuss its properties, including basis and dimension.

1.2 Vector Spaces

Why do we bother with a list of eight axioms? To a student first encountering Linear Algebra, this can feel like a bureaucratic exercise in “checking boxes.” However, there is a profound beauty hidden in this formality.

In our preliminary chapter, we saw three very different worlds: the world of geometric arrows ($\mathbb{R}^n$), the world of functions ($F[x]$), and the world of matrices ($M_{m \times n}$). On the surface, an arrow is not a polynomial, and a polynomial is not a matrix. But if you squint, they all behave the same: you can add them, and you can scale them.

By defining a **Vector Space** through these axioms, we are choosing to ignore what the objects *are* and focus entirely on how they *act*. If we prove a theorem using only these eight axioms, that theorem becomes a “universal law” that applies to arrows, functions, and matrices all at once. This is the power of abstraction: solve the problem once, and you solve it for every universe that obeys these rules.

1.2.1 Vector Spaces and Vectors

Definition 1.2.1 — Vector Space

A set V over a field F with two operations: addition $+$ and scalar multiplication $\cdot$.

$$+ : V \times V \rightarrow V, \quad \cdot : F \times V \rightarrow V$$

such that satisfy the following axioms for all $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$ and $\alpha, \beta \in F$:

1. $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$
2. $(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 = \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)$
3. There exists $\mathbf{0} \in V$ such that $\mathbf{v}_1 + \mathbf{0} = \mathbf{v}_1$

4. There exists $\mathbf{v}'_1 \in V$ such that $\mathbf{v}_1 + \mathbf{v}'_1 = \mathbf{0}$
5. $\alpha(\mathbf{v}_1 + \mathbf{v}_2) = \alpha\mathbf{v}_1 + \alpha\mathbf{v}_2$
6. $(\alpha + \beta)\mathbf{v}_1 = \alpha\mathbf{v}_1 + \beta\mathbf{v}_1$
7. $\alpha(\beta\mathbf{v}_1) = (\alpha\beta)\mathbf{v}_1$
8. $1\mathbf{v}_1 = \mathbf{v}_1$

Remark. Instead of writing $(V, +, \cdot)$ is a vector space over F , we usually simplify it to V is a vector space over F , or even just V is a vector space if the context is clear.

Remark. We sometimes will abuse the name and refer to $\mathbf{0} = \mathbf{0}_V \in V$ as the “zero vector” of the vector space V . Readers should not confuse $\mathbf{0}_V$ with the zero scalar $0 = 0_F \in F$ from the field F .

Example. (Examples of Vector Spaces)

The following are classic examples of vector spaces:

1. Most defaultly, F^n with the usual operations of $+$ and $\cdot$ is a vector space over F .
2. Working with polynomials is common too: $F[x]$ with the usual operations of $+$ and $\cdot$ is a vector space over F .
3. Let $\mathcal{D}$ be any open interval. Let

$$C(\mathcal{D}) := \{f : \mathcal{D} \rightarrow \mathbb{R} : f \text{ is continuous}\}.$$

Then $C(\mathcal{D})$ is a vector space over $\mathbb{R}$. The zero vector of this vector space is given by the zero function ($x \mapsto 0$).

4. Let F be any field and fix $n \in \mathbb{N}$. Then $M_n(F)$ is a vector space over F .
5. The set $V = \mathbb{R}_{>0}$ of positive real numbers forms a vector space over $F = \mathbb{R}$ under the following operations: for $x, y \in V$ and $\alpha \in F$, define

$$x \oplus y := xy, \quad \alpha \odot x := x^\alpha = e^{\alpha \log x}.$$

Under these operations, the zero vector is $\mathbf{0} = 1$, and the additive inverse of x is x^{-1} .

You can check if addition and scalar multiplication make sense and follow all axioms of vector spaces.

Example. ($\mathbb{Q}$ Adjoin $\sqrt{2}$)

Show that $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a vector space over $\mathbb{Q}$ by checking all eight axioms.

Solution. Let $\mathbf{v}_1 = a_1 + b_1\sqrt{2}$, $\mathbf{v}_2 = a_2 + b_2\sqrt{2}$, $\mathbf{v}_3 = a_3 + b_3\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ and $\alpha, \beta \in \mathbb{Q}$.

1.

$$\begin{aligned}
\mathbf{v}_1 + \mathbf{v}_2 &= (a_1 + a_2) + (b_1 + b_2)\sqrt{2} \\
&= (a_2 + a_1) + (b_2 + b_1)\sqrt{2} \\
&= \mathbf{v}_2 + \mathbf{v}_1
\end{aligned}$$

2.

$$\begin{aligned}
(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 &= (a_1 + a_2 + a_3) + (b_1 + b_2 + b_3)\sqrt{2} \\
&= \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)
\end{aligned}$$

3. The element $\mathbf{0} = 0 + 0\sqrt{2} = 0 \in \mathbb{Q}(\sqrt{2})$ satisfies

$$\begin{aligned}
\mathbf{v}_1 + \mathbf{0} &= (a_1 + 0) + (b_1 + 0)\sqrt{2} \\
&= \mathbf{v}_1
\end{aligned}$$

4. For $\mathbf{v}_1 = a_1 + b_1\sqrt{2}$, define $-\mathbf{v}_1 := (-a_1) + (-b_1)\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. Then

$$\begin{aligned}
\mathbf{v}_1 + (-\mathbf{v}_1) &= (a_1 - a_1) + (b_1 - b_1)\sqrt{2} \\
&= \mathbf{0}
\end{aligned}$$

5.

$$\begin{aligned}
\alpha(\mathbf{v}_1 + \mathbf{v}_2) &= \alpha(a_1 + a_2) + \alpha(b_1 + b_2)\sqrt{2} \\
&= (\alpha a_1 + \alpha a_2) + (\alpha b_1 + \alpha b_2)\sqrt{2} \\
&= \alpha \mathbf{v}_1 + \alpha \mathbf{v}_2
\end{aligned}$$

6.

$$\begin{aligned}
(\alpha + \beta)\mathbf{v}_1 &= (\alpha + \beta)a_1 + (\alpha + \beta)b_1\sqrt{2} \\
&= (\alpha a_1 + \beta a_1) + (\alpha b_1 + \beta b_1)\sqrt{2} \\
&= \alpha \mathbf{v}_1 + \beta \mathbf{v}_1
\end{aligned}$$

7.

$$\begin{aligned}
\alpha(\beta \mathbf{v}_1) &= \alpha(\beta a_1 + \beta b_1\sqrt{2}) \\
&= \alpha\beta a_1 + \alpha\beta b_1\sqrt{2} \\
&= (\alpha\beta)\mathbf{v}_1
\end{aligned}$$

8.

$$\begin{aligned}
1 \cdot \mathbf{v}_1 &= 1 \cdot a_1 + 1 \cdot b_1\sqrt{2} \\
&= a_1 + b_1\sqrt{2} \\
&= \mathbf{v}_1
\end{aligned}$$

Thus $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}$.**Definition 1.2.2 — Vectors and Scalars**

Let V be a vector space over F . Then the elements of V are called **vectors**, and the elements of F are called **scalars**.

Additionally, $\mathbf{0} \in V$ is called the **zero vector**, and $(\mathbf{v}'_1)$ in (VS4) is called the **inverse element** of $\mathbf{v}_1$.

1.2.2 Vector Space Theorems

Theorem 1.2.1 — Left Cancellation Law

Let V be a vector space over F , let $\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V$. If $\mathbf{u} + \mathbf{v}_1 = \mathbf{u} + \mathbf{v}_2$, then $\mathbf{v}_1 = \mathbf{v}_2$.

Proof. Let $\mathbf{u}'$ be an inverse of $\mathbf{u}$.

$$\begin{aligned}
 \mathbf{u} + \mathbf{v}_1 &= \mathbf{u} + \mathbf{v}_2 && \text{(given)} \\
 \mathbf{u}' + (\mathbf{u} + \mathbf{v}_1) &= \mathbf{u}' + (\mathbf{u} + \mathbf{v}_2) && \text{(add } \mathbf{u}' \text{ to both sides)} \\
 (\mathbf{u}' + \mathbf{u}) + \mathbf{v}_1 &= (\mathbf{u}' + \mathbf{u}) + \mathbf{v}_2 && \text{(by VS2)} \\
 \mathbf{0} + \mathbf{v}_1 &= \mathbf{0} + \mathbf{v}_2 && \text{(by VS4)} \\
 \mathbf{v}_1 &= \mathbf{v}_2 && \text{(by VS3)}
 \end{aligned}$$

■

Theorem 1.2.2 — Right Cancellation Law

Let V be a vector space over F , let $\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V$. If $\mathbf{v}_1 + \mathbf{u} = \mathbf{v}_2 + \mathbf{u}$, then $\mathbf{v}_1 = \mathbf{v}_2$.

Proof. Since we have $\mathbf{u} + \mathbf{v}_1 = \mathbf{u} + \mathbf{v}_2 \implies \mathbf{v}_1 = \mathbf{v}_2$, applying (VS1) to it gives $\mathbf{v}_1 + \mathbf{u} = \mathbf{v}_2 + \mathbf{u} \implies \mathbf{v}_1 = \mathbf{v}_2$.

■

Theorem 1.2.3 — Zero Vector is Unique

Let V be a vector space over F . The zero vector $\mathbf{0} \in V$ is unique.

Proof. Suppose that there are two vectors $\mathbf{0}_1, \mathbf{0}_2$.

$$\begin{aligned}
 \mathbf{0}_1 + \mathbf{0}_2 &= \mathbf{0}_2 + \mathbf{0}_1 && \text{(by VS1)} \\
 \mathbf{0}_1 &= \mathbf{0}_2 && \text{(by VS3)}
 \end{aligned}$$

■

Theorem 1.2.4 — Additive Inverse is Unique

Let V be a vector space over F . Then for every $\mathbf{v} \in V$, its additive inverse described in (VS4), which is $\mathbf{v}'$, is unique.

Proof. Let $\mathbf{v}'_1, \mathbf{v}'_2$ both be inverses of $\mathbf{v}$. Then

$$\begin{aligned}
 \mathbf{v}'_1 &= \mathbf{v}'_1 + \mathbf{0} && \text{(by VS3)} \\
 &= \mathbf{v}'_1 + (\mathbf{v} + \mathbf{v}'_2) && \text{(by VS4)} \\
 &= (\mathbf{v}'_1 + \mathbf{v}) + \mathbf{v}'_2 && \text{(by VS2)} \\
 &= \mathbf{0} + \mathbf{v}'_2 && \text{(by VS4)} \\
 &= \mathbf{v}'_2 && \text{(by VS3)}
 \end{aligned}$$

■

Definition 1.2.3 — Notation of Additive Inverse

The unique inverse of a vector $\mathbf{v} \in V$, for a vector space V over F , will be denoted as $-\mathbf{v}$.

Theorem 1.2.5 — Zero Scalar Annihilates

Let $\mathbf{v} \in V$ for a vector space V over F . Then $0 \cdot \mathbf{v} = \mathbf{0}$.

Proof. Let $\mathbf{w} = 0 \cdot \mathbf{v}$. We show that $\mathbf{w} = \mathbf{0}$.

$$\begin{aligned}
 \mathbf{w} + \mathbf{w} &= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} \\
 &= (0 + 0) \cdot \mathbf{v} && \text{(by VS6)} \\
 &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\
 &= \mathbf{w} \\
 (\mathbf{w} + \mathbf{w}) + (-\mathbf{w}) &= \mathbf{w} + (-\mathbf{w}) && \text{(add } -\mathbf{w} \text{ to both sides)} \\
 \mathbf{w} + (\mathbf{w} + (-\mathbf{w})) &= \mathbf{0} && \text{(by VS2, VS4)} \\
 \mathbf{w} + \mathbf{0} &= \mathbf{0} && \text{(by VS4)} \\
 \mathbf{w} &= \mathbf{0} && \text{(by VS3)}
 \end{aligned}$$

■

Theorem 1.2.6 — Negation as Scalar Multiplication

Let V be a vector space over F and $\mathbf{v} \in V$ be any vector. Then $-\mathbf{v} = (-1) \cdot \mathbf{v}$.

Proof. We show that $(-1) \cdot \mathbf{v}$ is an additive inverse of $\mathbf{v}$:

$$\begin{aligned} \mathbf{v} + (-1) \cdot \mathbf{v} &= 1 \cdot \mathbf{v} + (-1) \cdot \mathbf{v} && \text{(by VS8)} \\ &= (1 + (-1)) \cdot \mathbf{v} && \text{(by VS6)} \\ &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\ &= \mathbf{0} && (*) \end{aligned}$$

By (*), we use 1.2.5.

Since $(-1) \cdot \mathbf{v}$ is an additive inverse of $\mathbf{v}$, by uniqueness we have $-\mathbf{v} = (-1) \cdot \mathbf{v}$. ■

Theorem 1.2.7 — Negative Scalar Distribution

Let V be a vector space over F . For any $\alpha \in F$ and $\mathbf{v} \in V$, we have $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$.

Proof. We show that $(-\alpha)\mathbf{v}$ is an additive inverse of $\alpha\mathbf{v}$:

$$\begin{aligned} \alpha\mathbf{v} + (-\alpha)\mathbf{v} &= (\alpha + (-\alpha))\mathbf{v} && \text{(by VS6)} \\ &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\ &= \mathbf{0} && (*) \end{aligned}$$

By (*), we use 1.2.5.

Since $(-\alpha)\mathbf{v}$ is an additive inverse of $\alpha\mathbf{v}$, by uniqueness we have $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$. ■

Theorem 1.2.8 — Scalar Multiplication by Zero Vector

Let V be a vector space over F . For any scalar $\alpha \in F$, we have $\alpha\mathbf{0} = \mathbf{0}$.

Proof. Let $\mathbf{w} = \alpha\mathbf{0}$. We show that $\mathbf{w} = \mathbf{0}$.

$$\begin{aligned} \mathbf{w} &= \alpha\mathbf{0} \\ &= \alpha(\mathbf{0} + \mathbf{0}) && \text{(by VS3)} \\ &= \alpha\mathbf{0} + \alpha\mathbf{0} && \text{(by VS5)} \\ &= \mathbf{w} + \mathbf{w} \\ \mathbf{w} + (-\mathbf{w}) &= (\mathbf{w} + \mathbf{w}) + (-\mathbf{w}) && \text{(add } -\mathbf{w} \text{ to both sides)} \\ \mathbf{0} &= \mathbf{w} + (\mathbf{w} + (-\mathbf{w})) && \text{(by VS4, VS2)} \\ \mathbf{0} &= \mathbf{w} + \mathbf{0} && \text{(by VS4)} \\ \mathbf{0} &= \mathbf{w} && \text{(by VS3)} \end{aligned}$$

■

Theorem 1.2.9 — Zero Product Law

Let V be a vector space over F , $\mathbf{v} \in V$, and $\alpha \in F$. If $\alpha\mathbf{v} = \mathbf{0}$, then either $\alpha = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. Suppose $\alpha\mathbf{v} = \mathbf{0}$. We consider two cases:

- **Case $\alpha = 0$:** Then the statement holds.
- **Case $\alpha \neq 0$:** Since F is a field, α has a multiplicative inverse α^{-1} .

$$\begin{array}{ll}
 \alpha\mathbf{v} = \mathbf{0} & \text{(given)} \\
 \alpha^{-1}(\alpha\mathbf{v}) = \alpha^{-1}\mathbf{0} & \text{(multiply by } \alpha^{-1}\text{)} \\
 (\alpha^{-1}\alpha)\mathbf{v} = \mathbf{0} & \text{(by VS7 and previous Theorem)} \\
 1 \cdot \mathbf{v} = \mathbf{0} & \text{(field inverse property)} \\
 \mathbf{v} = \mathbf{0} & \text{(by VS8)}
 \end{array}$$

Thus, if $\alpha \neq 0$, we must have $\mathbf{v} = \mathbf{0}$. ■

1.3 Subspaces

Some vector spaces might be “too big” and be unwieldy to deal with. Therefore, it is often convenient to look at “smaller vector spaces” inside a given vector space. These are called **subspaces**.

1.3.1 Definition and the Subspace Test

Definition 1.3.1 — Subspace

Let V be a vector space over F . A subset W of V is called a **subspace** of V , if W is also a vector space over F with addition and scalar multiplication inherited from V .

Remark. Let $(V, +, \cdot)$ be a vector space over F , and $W \subseteq V$. If $(W, +, \cdot)$ is also a vector space over F then W is a subspace of V .

Example. (Trivial Subspaces)

Given a vector space V , then

- V is a subspace of V itself;
- $\{\mathbf{0}\}$ is a subspace of V (also known as the “zero subspace”);
- $\emptyset$ is **not** a subspace of V .

Theorem 1.3.1 — Subspace Test

Let V be a vector space over F and $W \subseteq V$. Then W is a subspace of V if and only if the following are satisfied:

- W is non-empty;
- W is closed under addition: $\mathbf{w}_1 + \mathbf{w}_2 \in W$ for all $\mathbf{w}_1, \mathbf{w}_2 \in W$;
- W is closed under scalar multiplication: $\alpha \mathbf{w} \in W$ for all $\alpha \in F$ and $\mathbf{w} \in W$.

Proof. ($\Rightarrow$) If W is a subspace, then W is a vector space, so it contains the zero vector (hence non-empty) and is closed under addition and scalar multiplication by definition.

($\Leftarrow$) Assume the three conditions hold. Since $W \subseteq V$, the operations on W are inherited from V , so associativity, commutativity, and distributivity are automatically satisfied. It remains to verify the existence of identity elements and inverses.

- **Zero vector:** Since $W \neq \emptyset$, there exists $\mathbf{w} \in W$. By closure under scalar multiplication, $0 \cdot \mathbf{w} = \mathbf{0} \in W$.
- **Additive inverse:** For any $\mathbf{w} \in W$, closure under scalar multiplication gives $(-1) \cdot \mathbf{w} = -\mathbf{w} \in W$.

Thus W is a vector space under the inherited operations, i.e., a subspace of V . ■

Remark. Note that from the proof, the first condition of the Subspace Test can be replaced by the requirement that W contains the zero vector of V .

Remark. In practice, to test whether a given subset of a vector space is a subspace, we use 1.3.1 instead of checking all eight vector space axioms from scratch, which is much less tedious. Moreover, some introductory textbooks use 1.3.1 as the definition of a subspace. While this is logically equivalent, it can be somewhat less intuitive than defining a subspace as a subset that is itself a vector space.

1.3.2 Examples of Subspaces

Example. (Line as Subspace)

$W = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$ is a subspace of $\mathbb{R}^2$. Indeed:

- **Zero vector:** $(0, 0) \in W$ since $0 + 2(0) = 0$.
- **Addition:** If $(x_1, y_1), (x_2, y_2) \in W$, then $(x_1 + x_2) + 2(y_1 + y_2) = (x_1 + 2y_1) + (x_2 + 2y_2) = 0 + 0 = 0$, so their sum is in W .
- **Scalar multiplication:** If $(x, y) \in W$ and $c \in \mathbb{R}$, then $(cx) + 2(cy) = c(x + 2y) = c(0) = 0$, so $c(x, y) \in W$.

Example. (Translated Line is Not a Subspace)

$W = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$ is **not** a subspace of $\mathbb{R}^2$. This is because it does not contain the zero vector: $0 + 2(0) = 0 \neq 3$, so $(0, 0) \notin W$.

Example. (Polynomials of Bounded Degree)

Let $F[x]$ be the vector space of all polynomials over a field F . For a fixed $n \in \mathbb{N}$, the set $F[x]_{\leq n}$ of polynomials of degree at most n is a subspace of $F[x]$.

- **Zero vector:** The zero polynomial has degree $-\infty$, so $0 \in F[x]_{\leq n}$.
- **Addition:** If $\deg p \leq n$ and $\deg q \leq n$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$.
- **Scalar multiplication:** If $\deg p \leq n$ and $c \in F$, then $\deg(cp) \leq \deg p \leq n$.

Example. (Polynomials of Exact Degree)

The set $F[x]_{=n}$ of polynomials of degree **exactly** n is **not** a subspace of $F[x]$.

- It does not contain the zero vector (since $\deg 0 = -\infty \neq n$).
- It is not closed under addition. For example, if $p(x) = x^n + x$ and $q(x) = -x^n$, both have degree n , but $p(x) + q(x) = x$, which has degree $1 \neq n$ (assuming $n > 1$).

Example. (Differentiable Functions)

Let $C(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ be the set of all continuous functions on $\mathbb{R}$. We consider the set of differentiable functions:

$$D(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is differentiable}\}$$

To verify that $D(\mathbb{R})$ is a subspace of $C(\mathbb{R})$:

- **Subset:** Since differentiability implies continuity, we have $D(\mathbb{R}) \subseteq C(\mathbb{R})$.
- **Zero vector:** The zero function $f(x) = 0$ is differentiable, so $\mathbf{0} \in D(\mathbb{R})$.
- **Closure under addition:** If $f, g \in D(\mathbb{R})$, then $f + g$ is differentiable with $(f + g)' = f' + g'$, so $f + g \in D(\mathbb{R})$.
- **Closure under scalar multiplication:** If $f \in D(\mathbb{R})$ and $c \in \mathbb{R}$, then cf is differentiable with $(cf)' = cf'$, so $cf \in D(\mathbb{R})$.

Thus, $D(\mathbb{R})$ is a subspace of $C(\mathbb{R})$.

Example. (Symmetric Matrices)

The set of all symmetric matrices in $M_n(F)$, denoted by $S_n(F) = \{\mathbf{A} \in M_n(F) : \mathbf{A} = \mathbf{A}^\top\}$, is a subspace of $M_n(F)$.

- **Zero vector:** The zero matrix $\mathbf{O}$ is symmetric since $\mathbf{O}^\top = \mathbf{O}$, so $\mathbf{O} \in S_n(F)$.
- **Addition:** If $\mathbf{A}, \mathbf{B} \in S_n(F)$, then $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top = \mathbf{A} + \mathbf{B}$, so $\mathbf{A} + \mathbf{B} \in S_n(F)$.
- **Scalar multiplication:** If $\mathbf{A} \in S_n(F)$ and $c \in F$, then $(c\mathbf{A})^\top = c\mathbf{A}^\top = c\mathbf{A}$, so $c\mathbf{A} \in S_n(F)$.

Example. (Trace-free Matrices)

The set of all trace-free matrices in $M_n(F)$, denoted by $W = \{\mathbf{A} \in M_n(F) : \text{tr}(\mathbf{A}) = 0\}$, is a subspace of $M_n(F)$.

- **Zero vector:** $\text{tr}(\mathbf{O}) = 0 + 0 + \cdots + 0 = 0$, so $\mathbf{O} \in W$.
- **Addition:** If $\mathbf{A}, \mathbf{B} \in W$, then $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}) = 0 + 0 = 0$, so $\mathbf{A} + \mathbf{B} \in W$.
- **Scalar multiplication:** If $\mathbf{A} \in W$ and $c \in F$, then $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A}) = c(0) = 0$, so $c\mathbf{A} \in W$.

1.3.3 Intersections and Unions

Theorem 1.3.2 — Intersection of Subspaces

Let W_1 and W_2 be subspaces of a vector space V . Then the intersection $W_1 \cap W_2$ is also a subspace of V .

Proof. We use the Subspace Test on $W = W_1 \cap W_2$:

- **Zero vector:** Since W_1 and W_2 are subspaces, $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$. Thus $\mathbf{0} \in W_1 \cap W_2$.
- **Closure under addition:** Let $\mathbf{w}, \mathbf{z} \in W_1 \cap W_2$. Then $\mathbf{w}, \mathbf{z} \in W_1$ and $\mathbf{w}, \mathbf{z} \in W_2$. Since W_1 and W_2 are subspaces, they are closed under addition, so $\mathbf{w} + \mathbf{z} \in W_1$ and $\mathbf{w} + \mathbf{z} \in W_2$. Hence $\mathbf{w} + \mathbf{z} \in W_1 \cap W_2$.
- **Closure under scalar multiplication:** Let $\mathbf{w} \in W_1 \cap W_2$ and $c \in F$. Then $\mathbf{w} \in W_1$ and $\mathbf{w} \in W_2$. Since W_1 and W_2 are closed under scalar multiplication, $c\mathbf{w} \in W_1$ and $c\mathbf{w} \in W_2$. Hence $c\mathbf{w} \in W_1 \cap W_2$.

Since all three conditions of the Subspace Test are satisfied, $W_1 \cap W_2$ is a subspace of V . ■

Remark. Unlike the intersection, the **union** of two subspaces is not necessarily a subspace. For $W_1 \cup W_2$ to be a subspace, one subspace must be contained within the other (i.e., $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$).

Example. (Union of Axes)

Consider the vector space $V = \mathbb{R}^2$. Let W_1 be the x -axis and W_2 be the y -axis:

$$W_1 = \{(x, 0) : x \in \mathbb{R}\}, \quad W_2 = \{(0, y) : y \in \mathbb{R}\}.$$

Both W_1 and W_2 are subspaces of $\mathbb{R}^2$. However, their union $W_1 \cup W_2$ is **not** a subspace because it is not closed under addition.

Indeed, $(1, 0) \in W_1 \subseteq W_1 \cup W_2$ and $(0, 1) \in W_2 \subseteq W_1 \cup W_2$, but their sum:

$$(1, 0) + (0, 1) = (1, 1)$$

is not in $W_1 \cup W_2$ since $(1, 1)$ is neither on the x -axis nor the y -axis.

Proposition 1.3.3 — Union of Subspaces

Let W_1, W_2 be subspaces of V . Then $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Proof. ($\Leftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, which is a subspace. Similarly, if $W_2 \subseteq W_1$, the union is W_1 , which is also a subspace.

($\Rightarrow$) We prove the contrapositive: if neither is contained in the other, then the union is not a subspace. Assume $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$.

- Since $W_1 \not\subseteq W_2$, there exists a vector $\mathbf{u} \in W_1$ such that $\mathbf{u} \notin W_2$.
- Since $W_2 \not\subseteq W_1$, there exists a vector $\mathbf{v} \in W_2$ such that $\mathbf{v} \notin W_1$.

Consider the sum $\mathbf{w} = \mathbf{u} + \mathbf{v}$. We show that $\mathbf{w} \notin W_1 \cup W_2$:

- If $\mathbf{w} \in W_1$, then $\mathbf{v} = \mathbf{w} - \mathbf{u}$. Since $\mathbf{w} \in W_1$ and $\mathbf{u} \in W_1$, their difference $\mathbf{v}$ must be in W_1 (by closure). But we chose $\mathbf{v} \notin W_1$, which is a contradiction.
- If $\mathbf{w} \in W_2$, then $\mathbf{u} = \mathbf{w} - \mathbf{v}$. Since $\mathbf{w} \in W_2$ and $\mathbf{v} \in W_2$, their difference $\mathbf{u}$ must be in W_2 . But we chose $\mathbf{u} \notin W_2$, which is a contradiction.

Therefore, $\mathbf{u} + \mathbf{v}$ is in neither W_1 nor W_2 , so it is not in $W_1 \cup W_2$. Thus, the union is not closed under addition and is therefore not a subspace. ■

1.4 Linear Combinations and Span

In the previous section, we studied subspaces. A natural question arises: given a set of vectors, how can we describe the “smallest” subspace that contains them? This leads us to the concept of linear combinations and the span.

1.4.1 Linear Combinations

Definition 1.4.1 — Linear Combination

Let V be a vector space over F , $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then the expression

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m$$

is called a **linear combination** of S . The scalars a_i are called the **coefficients** of the

linear combination.

Example. (Examples of Linear Combinations)

Below are some examples of linear combinations in different vector spaces:

1. The vector $(2, 1, 0)$ in $\mathbb{R}^3$ is a linear combination of $S = \{(1, 2, 3), (4, 5, 6)\}$ because

$$(2, 1, 0) = -2 \cdot (1, 2, 3) + 1 \cdot (4, 5, 6).$$

2. The polynomial $f(x) = x^3 + 2x + 1$ in $\mathbb{R}[x]$ is a linear combination of $S = \{1, x, x^2, x^3\}$ because

$$f(x) = 1 \cdot 1 + 2 \cdot x + 0 \cdot x^2 + 1 \cdot x^3.$$

1.4.2 Span and Spanning Sets

Definition 1.4.2 — Span

Let S be a subset of a vector space V over F . The **span** of S , denoted by $\text{span}_F(S)$, is the set of all linear combinations of finite subsets of elements of S .

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then $\text{span}_F(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n : a_i \in F\}$. By convention, $\text{span}_F(\emptyset) = \{\mathbf{0}\}$.

Remark. When the underlying field is clear, we abuse the notation of span_F and write it as span .

Theorem 1.4.1 — Spanning Set as Subspace

The span of any subset $S \subseteq V$ is a subspace of V . Moreover, it is the smallest subspace of V containing S .

Proof. Let $W = \text{span}(S)$.

- **Zero vector:** Since $\text{span}(\emptyset) = \{\mathbf{0}\}$, and for non-empty S , taking all coefficients as zero gives $\mathbf{0}$, we have $\mathbf{0} \in W$.
- **Addition:** Let $\mathbf{x}, \mathbf{y} \in W$. Then $\mathbf{x} = \sum a_i \mathbf{v}_i$ and $\mathbf{y} = \sum b_i \mathbf{v}_i$. Their sum $\mathbf{x} + \mathbf{y} = \sum (a_i + b_i) \mathbf{v}_i$ is also a linear combination of elements in S , so $\mathbf{x} + \mathbf{y} \in W$.
- **Scalar multiplication:** For any $c \in F$, $c\mathbf{x} = \sum (ca_i) \mathbf{v}_i \in W$.

To see it is the smallest subspace, note that any subspace containing S must be closed under addition and scalar multiplication, and thus must contain all linear combinations of elements in S .

■

Definition 1.4.3 — Spanning Set

If $\text{span}(S) = V$, we say that S **spans** V , or that S is a **spanning set** for V .

Example. (Spanning Set for Polynomials)

The set $\{1, x, x^2\}$ spans $\mathbb{R}[x]_{\leq 2}$, the space of polynomials of degree at most 2.

1.5 Linear Independence

A spanning set is like a toolbox that is guaranteed to have every tool you need. But a messy toolbox might have three different hammers that all do the same thing. In mathematics, as in engineering, we value **efficiency**.

If one of the vectors in your spanning set can already be built using the others, then that vector is “dead weight”—it isn’t helping you reach any new territory. We call such a set *linearly dependent*.

In this section, we search for the “cleanest” possible sets. We want to know: “Is every vector in this collection actually contributing something unique, or is someone just coasting on the work of the others?”

1.5.1 Definition and Examples

Definition 1.5.1 — Linear Independence

A subset S of a vector space V is said to be **linearly independent** if for any distinct vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in S$, the only solution to the equation:

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$$

is the trivial solution $a_1 = a_2 = \cdots = a_n = 0$.

If there exists a non-trivial solution (where at least one $a_i \neq 0$), then S is said to be **linearly dependent**.

Remark. Geometrically, in $\mathbb{R}^2$, two vectors are linearly dependent if and only if they lie on the same line through the origin (one is a multiple of the other). In $\mathbb{R}^3$, three vectors are linearly dependent if and only if they lie on the same plane through the origin.

Example. (Examples of Linear Independence)

The following examples illustrate how to check for linear independence in various settings:

1. Let $S_1 = \{(1, 2, 3), (4, 5, 6)\} \subseteq \mathbb{R}^3$. To check linear independence, suppose

$$a(1, 2, 3) + b(4, 5, 6) = (0, 0, 0) \implies \begin{cases} a + 4b = 0 \\ 2a + 5b = 0 \\ 3a + 6b = 0 \end{cases}$$

The first two equations give $a = -4b$ and $2(-4b) + 5b = 0 \implies -3b = 0$. Thus $a = b = 0$, so S_1 is **linearly independent**.

2. Let $S_2 = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\} \subseteq \mathbb{R}^3$. Note that

$$(1, 2, 3) - 2(4, 5, 6) + (7, 8, 9) = (1 - 8 + 7, 2 - 10 + 8, 3 - 12 + 9) = (0, 0, 0).$$

Since there exists a non-trivial solution, S_2 is **linearly dependent**.

3. Consider $\{2x + 1, x^2 + 3\}$ in $\mathbb{R}[x]_{\leq 2}$. Suppose

$$a(2x + 1) + b(x^2 + 3) = 0 \implies bx^2 + 2ax + (a + 3b) = 0.$$

Comparing coefficients of $x^2, x, 1$, we have $b = 0$, $2a = 0$, and $a + 3b = 0$. This implies $a = b = 0$, so the set is **linearly independent**.

4. Consider $\{\sin x, \cos x, e^x\}$ in $D(\mathbb{R})$. Suppose for all $x \in \mathbb{R}$:

$$a \sin x + b \cos x + ce^x = 0.$$

Setting $x = 0$ gives $b + c = 0$. Setting $x = \pi$ gives $-b + ce^\pi = 0$. Adding these gives $c(1 + e^\pi) = 0$, so $c = 0$ and hence $b = 0$. Finally, setting $x = \pi/2$ gives $a + ce^{\pi/2} = 0 \implies a = 0$. Thus, the set is **linearly independent**.

Theorem 1.5.1 — Linear Dependence Lemma

$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent if and only if either $\mathbf{v}_1 = \mathbf{0}$ or for some r , $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}\}$.

Proof. ($\implies$) Suppose $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly dependent. Then there exist $a_1, \dots, a_m \in F$, not all zero, such that

$$a_1 \mathbf{v}_1 + \dots + a_m \mathbf{v}_m = \mathbf{0}.$$

Let r be the largest index such that $a_r \neq 0$. Then

$$a_1 \mathbf{v}_1 + \dots + a_{r-1} \mathbf{v}_{r-1} + a_r \mathbf{v}_r = \mathbf{0}.$$

So

$$a_r \mathbf{v}_r = -(a_1 \mathbf{v}_1 + \dots + a_{r-1} \mathbf{v}_{r-1}),$$

and dividing by a_r gives

$$\mathbf{v}_r = -\frac{a_1}{a_r} \mathbf{v}_1 - \dots - \frac{a_{r-1}}{a_r} \mathbf{v}_{r-1}.$$

Hence $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$. (If $r = 1$, this says $a_1 \mathbf{v}_1 = \mathbf{0}$ with $a_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$.)

($\impliedby$) We prove by cases:

- If $\mathbf{v}_1 = \mathbf{0}$, then

$$1 \cdot \mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + \cdots + 0 \cdot \mathbf{v}_m = \mathbf{0},$$

with coefficients not all zero, so S is linearly dependent.

- If for some r we have $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$, then there exist scalars $c_1, \dots, c_{r-1} \in F$ such that

$$\mathbf{v}_r = c_1 \mathbf{v}_1 + \cdots + c_{r-1} \mathbf{v}_{r-1}.$$

Move everything to one side:

$$c_1 \mathbf{v}_1 + \cdots + c_{r-1} \mathbf{v}_{r-1} - 1 \cdot \mathbf{v}_r = \mathbf{0}.$$

Extending coefficients by zeros for $\mathbf{v}_{r+1}, \dots, \mathbf{v}_m$, we get a nontrivial linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$ equal to $\mathbf{0}$, so S is linearly dependent. ■

Remark. The above theorem can be paraphrased to: A set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ with $n \geq 2$ is linearly dependent if and only if at least one vector in S can be written as a linear combination of the others.

1.5.2 The Redundant Members

Having the last remark in mind, we can develop an intuition that says linear independence of a set is equivalent to requiring the set containing no “redundant members.” Next we want to develop the intuition saying that throwing away these “redundant members” will not change the linear span.

Theorem 1.5.2 — Span Preservation

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{w}\} \subseteq V$. If $\mathbf{w} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$, then

$$\text{span}(S) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}.$$

Proof. Let $W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $W' = \text{span}(S)$. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_m\} \subseteq S$, it is clear that $W \subseteq W'$.

For the reverse inclusion, let $\mathbf{v} \in W'$. Then $\mathbf{v}$ is a linear combination of vectors in S :

$$\mathbf{v} = a_1 \mathbf{v}_1 + \cdots + a_m \mathbf{v}_m + b \mathbf{w}$$

for some scalars a_i, b . Since $\mathbf{w} \in W$, there exist scalars $c_1, \dots, c_m$ such that $\mathbf{w} = c_1 \mathbf{v}_1 + \cdots + c_m \mathbf{v}_m$. Substituting this into the expression for $\mathbf{v}$:

$$\mathbf{v} = a_1 \mathbf{v}_1 + \cdots + a_m \mathbf{v}_m + b(c_1 \mathbf{v}_1 + \cdots + c_m \mathbf{v}_m) = (a_1 + bc_1) \mathbf{v}_1 + \cdots + (a_m + bc_m) \mathbf{v}_m.$$

Thus $\mathbf{v}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$, so $\mathbf{v} \in W$. This shows $W' \subseteq W$. Hence $W' = W$. ■

1.6 Bases

We have reached what many consider the “Goldilocks Zone” of Linear Algebra.

If a set is too small, it won’t span the whole space—you’ll be left with points you can’t reach. If a set is too large, it will be linearly dependent—you’ll have redundant vectors cluttering your workspace. A **basis** is a set that is “just right.” It is large enough to build everything, but small enough that every piece is essential.

Because a basis has no redundancy, it gives us something incredible: a **unique address system**. In an abstract vector space, it’s hard to tell someone where a vector is. But once we pick a basis, every vector can be described by a unique list of numbers—its coordinates. A basis is the bridge that allows us to turn abstract geometry into concrete arithmetic.

1.6.1 Definition and Standard Bases

Definition 1.6.1 — Basis

A subset B of a vector space V is called a **basis** for V if:

1. B is linearly independent;
2. B spans V .

Example. (Standard Bases)

Commonly used standard bases for familiar vector spaces are listed below:

- The **standard basis** for F^n is $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ has a 1 in the i -th position and 0 elsewhere.
- The standard basis for $F[x]_{\leq n}$ is $\{1, x, x^2, \dots, x^n\}$.
- The standard basis for $M_{m \times n}(F)$ is the set of matrices $\mathbf{E}_{ij}$ having a 1 at entry (i, j) and 0 elsewhere.

1.6.2 Unique Representation and Coordinates

The most important property of a basis is that every vector in the space has a unique “address” relative to it.

Theorem 1.6.1 — Unique Representation

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then every vector $\mathbf{v} \in V$ can be written as a linear combination of elements in B in **exactly one way**.

Proof. Since B spans V , there exist scalars a_i such that $\mathbf{v} = \sum a_i \mathbf{v}_i$. Suppose there is

another representation $\mathbf{v} = \sum b_i \mathbf{v}_i$. Then:

$$\mathbf{0} = \mathbf{v} - \mathbf{v} = \sum a_i \mathbf{v}_i - \sum b_i \mathbf{v}_i = \sum (a_i - b_i) \mathbf{v}_i.$$

Since B is linearly independent, we must have $a_i - b_i = 0$ for all i , meaning $a_i = b_i$. ■

Definition 1.6.2 — Coordinates

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an **ordered** basis for V . For any $\mathbf{v} \in V$, the unique scalars $c_1, \dots, c_n$ such that $\mathbf{v} = \sum c_i \mathbf{v}_i$ are called the **coordinates** of $\mathbf{v}$ with respect to B . We write this as a column vector:

$$[\mathbf{v}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

Remark. Notice the subtle but important distinction: a **basis as a set** determines which vectors can be built (i.e., it determines the space), but to define **coordinates** we need an **ordered basis**—a basis with a fixed ordering of its elements. For example, if $B = \{\mathbf{v}_1, \mathbf{v}_2\}$, then the coordinate vector $[\mathbf{v}]_B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ means $\mathbf{v} = 3\mathbf{v}_1 + 5\mathbf{v}_2$. If we had listed the basis in the opposite order $B' = \{\mathbf{v}_2, \mathbf{v}_1\}$, the same vector $\mathbf{v}$ would have coordinates $[\mathbf{v}]_{B'} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. The underlying space is the same, but the coordinate representation depends on the ordering.

Remark. The map $\mathbf{v} \mapsto [\mathbf{v}]_B$ provides a way to treat any abstract n -dimensional vector space as if it were simply F^n . This is the power of a basis.

1.7 Sifting and the Replacement Theorem

How “big” is a vector space? We intuitively know that a plane is “bigger” than a line, and 3D space is “bigger” than a plane. But in the abstract world of vector spaces, we need a rigorous way to measure this.

The answer lies in the number of vectors in a basis. But there’s a potential problem: what if one person finds a basis with 3 vectors, and another person finds a basis for the same space with 4 vectors? If that were possible, our entire concept of “size” would collapse.

In this section, we prove the most important technical result of the chapter: the **Steinitz Replacement Theorem**. This theorem is the “engine” that guarantees every basis of a space has the exact same number of vectors. Once we have this, we can finally define the **Dimension** of a space—a single number that captures its fundamental complexity.

1.7.1 The Sifting Method

Theorem 1.7.1 — Sifting Method

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then there exists a subset $S' \subseteq S$ such that:

1. S' is linearly independent, and
2. $\text{span}(S') = \text{span}(S)$.

Proof. For each $r = 1, 2, \dots, m$, we check whether $\mathbf{v}_r$ is a linear combination of the previous terms $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$. We:

1. Keep $\mathbf{v}_r$ in S' if $\mathbf{v}_r$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (not redundant);
2. Remove $\mathbf{v}_r$ from S' if $\mathbf{v}_r$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (redundant).

After this process, we obtain a subset S' in which no element is a linear combination of the preceding members anymore. Therefore, by 1.5.1, S' is linearly independent. Moreover, according to 1.5.2, removing “redundant members” from S does not change its linear span. Therefore, we have $\text{span}(S') = \text{span}(S)$. ■

Remark. The sifting method is a powerful algorithmic tool that allows us to “clean up” any set of vectors. Given any collection of vectors, we can systematically remove the redundant ones while preserving the span. This gives us a “cleaned” version that is linearly independent and spans the same space.

This method becomes our “new weapon” in linear algebra proofs. Whenever we encounter a set that might have redundant vectors, we can apply the sifting method to obtain a linearly independent subset with the same span. This technique will be particularly useful in proving results about dimension and in constructing bases from spanning sets.

Example. (Sifting Example)

Consider $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\}$ in $\mathbb{R}^3$ where:

$$\begin{array}{lll} \mathbf{v}_1 = (1, 1, 1), & \mathbf{v}_2 = (2, 2, 2), & \mathbf{v}_3 = (1, 0, 0), \\ \mathbf{v}_4 = (3, 2, 2), & \mathbf{v}_5 = (1, 1, 0), & \mathbf{v}_6 = (0, 0, 1). \end{array}$$

We apply the sifting method:

- $\mathbf{v}_1 \neq \mathbf{0}$, so $\mathbf{v}_1$ is kept.
- $\mathbf{v}_2 = 2\mathbf{v}_1$, so $\mathbf{v}_2$ is removed.
- $\mathbf{v}_3$ is not a linear combination of $\mathbf{v}_1$, so $\mathbf{v}_3$ is kept.
- $\mathbf{v}_4 = 2\mathbf{v}_1 + 1\mathbf{v}_3$, so $\mathbf{v}_4$ is removed.
- Suppose $\mathbf{v}_5 = a\mathbf{v}_1 + b\mathbf{v}_3$. Then $(1, 1, 0) = (a + b, a, a)$, which gives $a = 0$ and $a + b = 1$, so $b = 1$. But then the second coordinate gives $1 = a = 0$, which is a contradiction. So $\mathbf{v}_5$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_3$. Hence, $\mathbf{v}_5$ is kept.
- $\mathbf{v}_6 = (0, 0, 1) = (1, 1, 1) - (1, 1, 0) = \mathbf{v}_1 - \mathbf{v}_5$, so $\mathbf{v}_6$ is removed.

The sifted set is $S' = \{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5\} = \{(1, 1, 1), (1, 0, 0), (1, 1, 0)\}$, which is linearly independent and spans the same space as S .

Example. (Sifting in $\mathbb{R}^2$)

Consider $S = \{(1, 0), (0, 1), (1, 1)\}$ in $\mathbb{R}^2$.

- $(1, 0) \neq \mathbf{0}$, so $(1, 0)$ is kept.
- $(0, 1) \notin \text{span}\{(1, 0)\}$, so $(0, 1)$ is kept.
- $(1, 1) = (1, 0) + (0, 1) \in \text{span}\{(1, 0), (0, 1)\}$, so $(1, 1)$ is removed.

The sifted set $\{(1, 0), (0, 1)\}$ is linearly independent and still spans $\mathbb{R}^2$.

1.7.2 The Steinitz Replacement Theorem

Theorem 1.7.2 — Steinitz Exchange/Replacement Theorem

Let V be a vector space over F . Suppose:

- $G = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans V ;
- $L = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is linearly independent.

Then $m \leq n$.

Proof. The strategy is to insert the vectors from L one-by-one into G and “sift” the resulting set to maintain the spanning property. We rely on the fact that if we add a vector $\mathbf{w}$ to a spanning set S , the set $\{\mathbf{w}\} \cup S$ becomes linearly dependent, allowing us to remove a redundant vector (by 1.5.2).

Consider the first step. We form the set B_1 by inserting $\mathbf{u}_1$ into G :

$$B_1 = \{\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

Since G spans V , B_1 is linearly dependent. By applying the sifting process (1.7.1), we can remove a vector to obtain a new spanning set G_1 . Since L is linearly independent, $\mathbf{u}_1 \neq \mathbf{0}$, so the removed vector must be some $\mathbf{v}_i$. Thus:

$$G_1 = \{\mathbf{u}_1\} \cup S_1,$$

where S_1 is a subset of G with size $n - 1$.

Now, consider the general step. Suppose we have successfully exchanged k vectors such that the set

$$G_k = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \cup S_k$$

spans V , where $S_k \subset G$ contains $n - k$ vectors. We insert the next vector $\mathbf{u}_{k+1}$ to form:

$$B_{k+1} = \{\mathbf{u}_{k+1}\} \cup G_k.$$

Since G_k is a spanning set, $\mathbf{u}_{k+1}$ depends on the vectors in G_k , making B_{k+1} dependent. We must remove a vector to restore the spanning property. We cannot remove any of

the $\mathbf{u}_i$'s, as this would imply a linear dependence relation exclusively among elements of L , contradicting the hypothesis that L is linearly independent. Therefore, we must remove some $\mathbf{v}_j$ from S_k .

This process generates a sequence of subsets of remaining $\mathbf{v}$'s:

$$n = |G| > |S_1| > |S_2| > \cdots > |S_m| \geq 0.$$

Since we are able to perform this exchange m times (once for each $\mathbf{u} \in L$), we must have started with enough $\mathbf{v}$'s to accommodate every removal. Therefore, $n \geq m$. ■

1.7.3 Consequences of the Replacement Theorem

Having proved the Steinitz Replacement Theorem, the major conclusion we can draw from it is that any (finite) basis of the same vector space will have the same size.

Theorem 1.7.3 — Basis Theorem

If a vector space V over F has a finite basis, then every basis of V has the same number of vectors.

Proof. Let B_1 and B_2 be two bases with n and m vectors respectively.

- Since B_1 is linearly independent and B_2 spans, by Replacement Theorem, $n \leq m$.
- Since B_2 is linearly independent and B_1 spans, by Replacement Theorem, $m \leq n$.

Thus $n = m$. ■

Definition 1.7.1 — Dimension

A vector space V over F is called **finite-dimensional** if it has a finite basis. The **dimension** of V , denoted $\dim_F(V)$, is the number of vectors in any basis for V . By convention, $\dim_F(\{\mathbf{0}\}) = 0$.

Remark. The convention $\dim(\{\mathbf{0}\}) = 0$ is consistent with the fact that the **only basis for the zero space is the empty set** $\emptyset$. Indeed, $\emptyset$ is vacuously linearly independent (there is no nontrivial linear combination to consider), and $\text{span}(\emptyset) = \{\mathbf{0}\}$ by our earlier convention. Since $|\emptyset| = 0$, we have $\dim(\{\mathbf{0}\}) = 0$.

Remark. When the underlying field is clear, we abuse the notation of $\dim_F$ and write it as $\dim$.

Example. (Examples of Dimensions)

$\dim(F^n) = n$, $\dim(F[x]_{\leq n}) = n + 1$, and $\dim(M_{m \times n}(F)) = mn$.

Remark. The notation $\dim_F(V)$ reminds us that **dimension depends on the choice of field**. For instance, recall from 1.2.1 that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a vector space over $\mathbb{Q}$ with basis $\{1, \sqrt{2}\}$, so $\dim_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2})) = 2$.

As a more striking example, consider the complex numbers $\mathbb{C}$. As a vector space over itself, $\mathbb{C}$ has basis $\{1\}$, so $\dim_{\mathbb{C}}(\mathbb{C}) = 1$. However, as a vector space over $\mathbb{R}$, $\mathbb{C}$ has basis $\{1, i\}$, so $\dim_{\mathbb{R}}(\mathbb{C}) = 2$. The same set can have different dimensions depending on which field we use for scalar multiplication!

Theorem 1.7.4 — Size Bounds for Independent and Spanning Sets

Let V be a vector space of dimension n over F , and $S \subseteq V$ be a subset.

1. If $|S| > n$, then S cannot be linearly independent.
2. If $|S| < n$, then S cannot span V .

Proof.

1. Given $|S| > n$, and suppose for the sake of contradiction that S is linearly independent. Let B be a basis of V , then we have $|B| = n$ and B is a spanning set of V . By Replacement Theorem, we have

$$\underbrace{|S|}_{\text{size of lin. indep. set}} \leq \underbrace{|B|}_{\text{size of spanning set}} = n.$$

Giving a contradiction, hence S cannot be linearly independent.

2. Given $|S| < n$, and suppose for the sake of contradiction that S spans V . Let B be a basis of V , then we have $|B| = n$ and B is linearly independent. By Replacement Theorem, we have

$$n = \underbrace{|B|}_{\text{size of lin. indep. set}} \leq \underbrace{|S|}_{\text{size of spanning set}}.$$

This implies $n \leq |S|$, which contradicts the assumption that $|S| < n$. Hence S cannot span V . ■

Remark.

$$\text{Size of lin. indep. set} \leq \text{Size of basis} \leq \text{Size of spanning set}.$$

Theorem 1.7.5 — Basis Extension Theorem

Let V be a finite-dimensional vector space and S be a linearly independent subset of V .

Then S can be extended into a basis of V .

Proof. Since V is finite-dimensional, it has a finite basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Consider the set $S \cup B$. Since B spans V , $S \cup B$ also spans V . We apply the Sifting Method (1.7.1) to the ordered set formed by listing the elements of S first, followed by the elements of B .

The Sifting Method will keep all elements of S because S is linearly independent. It then proceeds to examine each element of B , keeping those that are not in the span of the vectors already kept. The resulting subset S' spans $\text{span}(S \cup B) = V$ and is linearly independent by construction. Thus, S' is a basis for V that contains S . ■

Corollary 1.7.6 — Existence of Basis

Every finite spanning set of a vector space V contains a basis.

Proof. Apply the Sifting Method to the spanning set. The resulting subset S' is linearly independent by construction and still spans V . Therefore, S' is a basis for V . ■

Theorem 1.7.7 — Dimension Implies Equality

Let V and W be finite-dimensional vector spaces over F . If

1. $W \subseteq V$;
2. $\dim_F(W) = \dim_F(V)$,

then $W = V$.

Proof. If we show $V \subseteq W$ then we are done. Suppose for contradiction that $V \not\subseteq W$. That means there exists some $\mathbf{v} \in V$ that $\mathbf{v} \notin W$.

Let $S = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be a basis of W , (consequently, $\dim_F(W) = n$). Since $\mathbf{v} \notin W$, we have $\mathbf{v} \notin \text{span}(S)$.

Consider $S' = \{\mathbf{w}_1, \dots, \mathbf{w}_n, \mathbf{v}\}$. Since S is linearly independent, no $\mathbf{w}_i$ is a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_{i-1}$; and $\mathbf{v}$ is not a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_n$ because $\mathbf{v} \notin \text{span}(S)$. By 1.5.1, S' is linearly independent in V . By Replacement Theorem, we have

$$\underbrace{|S'|}_{\text{size of lin. indep. set}} \leq \dim_F(V),$$

which implies $n + 1 \leq n$, which is a contradiction, therefore $V \subseteq W$. ■

Remark. This theorem makes proving $W = V$ somewhat easier. As if we wanted to prove equivalence before, we can only do it by proving $W \subseteq V$ and $V \subseteq W$. Now we can just prove one side and then state that their dimension is the same.

An immediate consequence of the above theorem is the following:

Theorem 1.7.8 — Maximal Independent Subset is a Basis

Let V be a vector space of dimension n over F . Then any n independent vectors form a basis of V . (We say that “any maximal linearly independent subset forms a basis”).

Proof. Let S be a linearly independent set of size n . We want to show that $\text{span}(S) = V$.

Let $W = \text{span}(S)$, this implies that W is a subspace of V , which implies $W \subseteq V$.

Notice that S spans W and S is linearly independent, therefore S is a basis of W . So $\dim_F(W) = |S| = n = \dim_F(V)$. By 1.7.7, we have $W = V$, then $\text{span}(S) = V$. ■

1.8 Sums of Subspaces

In the previous sections, we studied intersections and unions of subspaces. While the intersection of two subspaces is always a subspace, the union typically is not. This raises a natural question: what is the “right” way to combine two subspaces into a larger one?

The answer is the **sum** of subspaces. Given two subspaces W_1 and W_2 , their sum $W_1 + W_2$ consists of all vectors that can be written as $\mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. This construction always yields a subspace—in fact, it is the smallest subspace containing both W_1 and W_2 .

A particularly nice situation occurs when W_1 and W_2 have only the zero vector in common. In this case, their sum is called a **direct sum**, and every vector in the sum can be decomposed uniquely into a piece from W_1 and a piece from W_2 . Direct sums are fundamental in decomposing vector spaces into simpler, independent components.

1.8.1 Definition and Properties

Definition 1.8.1 — Sum of Subspaces

Let W_1, W_2 be subspaces of a vector space V . Then we define

$$W_1 + W_2 := \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2\}.$$

Theorem 1.8.1 — Sum of Subspaces Theorem

$W_1 + W_2$ is a subspace of V . Furthermore, this is the smallest subspace containing both W_1 and W_2 .

Proof. To prove that $W_1 + W_2$ is a subspace V . We prove that three conditions of the subspace test:

- Non-emptiness: As W_1, W_2 are subspaces, they both contains $\mathbf{0}$. And since $\mathbf{0} = \mathbf{0}_{W_1} + \mathbf{0}_{W_2}$, therefore $W_1 + W_2$ is non-empty.
- Closed under addition: Let $\mathbf{a}, \mathbf{b} \in W_1 + W_2$. Then we have

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2, & \text{for some } \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2 \\ \mathbf{b} = \mathbf{w}'_1 + \mathbf{w}'_2, & \text{for some } \mathbf{w}'_1 \in W_1, \mathbf{w}'_2 \in W_2 \end{cases}$$

Then $\mathbf{a} + \mathbf{b} = \underbrace{(\mathbf{w}_1 + \mathbf{w}'_1)}_{\in W_1} + \underbrace{(\mathbf{w}_2 + \mathbf{w}'_2)}_{\in W_2} \in W_1 + W_2$. So $W_1 + W_2$ is closed under addition.

- Closed under scalar multiplication: Let $\mathbf{a} \in W_1 + W_2$, and $\lambda \in F$. Then we have $\mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Then $\lambda \mathbf{a} = \underbrace{\lambda \mathbf{w}_1}_{\in W_1} + \underbrace{\lambda \mathbf{w}_2}_{\in W_2} \in W_1 + W_2$. So

$W_1 + W_2$ is closed under scalar multiplication.

This concludes that $W_1 + W_2$ is a subspace.

To prove that $W_1 + W_2$ is the smallest subspace containing both W_1 and W_2 , we first prove that it actually contains W_1 and W_2 , then we prove that it is the smallest subspace of V containing so.

- Containment: Since W_1 and W_2 are subspaces, they both contain the zero vector $\mathbf{0}$. Since for any $\mathbf{w}_1 \in W_1$, we can write $\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0}$, by $\mathbf{0} \in W_2$, we have $\mathbf{w}_1 \in W_1 + W_2$. Thus $W_1 \subseteq W_1 + W_2$. The same logic applies to W_2 therefore $W_2 \subseteq W_1 + W_2$.
- Minimality: Let U be any subspace of V such that $W_1 \subseteq U$ and $W_2 \subseteq U$. We want to show that $W_1 + W_2 \subseteq U$. Take any element $W_1 + W_2$. By definition $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. Because $W_1 \subseteq U$, then $\mathbf{w}_1 \in U$. Similarly, because $W_2 \subseteq U$, then $\mathbf{w}_2 \in U$. At last, since U is a subspace, it is closed under addition. Therefore

$$\mathbf{w}_1 + \mathbf{w}_2 \in U \implies \mathbf{v} \in U.$$

This concludes that $W_1 + W_2$ is the smallest subspace of V containing W_1 and W_2 . ■

1.8.2 Dimension of $W_1 + W_2$

Theorem 1.8.2 — Dimension Formula of Subspace Sum

Let V be a finite-dimensional vector space, and let W_1, W_2 subspaces of V . Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof. We use the above diagram to build intuition for the proof.

Let $n = \dim(W_1 \cap W_2)$. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the intersection $W_1 \cap W_2$.

Since B is a linearly independent subset of both W_1 and W_2 , by the Basis Extension Theorem (1.7.5), we can extend it to form bases for both subspaces:

1. **For** W_1 : Extend B with $\{\mathbf{s}_1, \dots, \mathbf{s}_p\}$ such that $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 . Note that $\dim(W_1) = n + p$.
2. **For** W_2 : Extend B with $\{\mathbf{t}_1, \dots, \mathbf{t}_q\}$ such that $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 . Note that $\dim(W_2) = n + q$.

Claim. The set $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of $W_1 + W_2$.

Proof.

- S spans $W_1 + W_2$
Let $\mathbf{w} \in W_1 + W_2$. By definition, $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Since B_1 spans W_1 and B_2 spans W_2 , both $\mathbf{w}_1$ and $\mathbf{w}_2$ can be written as linear combinations of vectors in S . Thus, their sum $\mathbf{w}$ is also in $\text{span}(S)$.
- S is Linearly Independent

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0} \quad (1.8.1)$$

Rearrange the equation to isolate the terms involving $\mathbf{t}_k$:

$$\mathbf{v} := \sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = - \sum_{k=1}^q \gamma_k \mathbf{t}_k$$

Observe the vector $\mathbf{v}$. From the LHS, we have $\mathbf{v} \in W_1$. From the RHS, we have $\mathbf{v} \in W_2$. Therefore, $\mathbf{v} \in W_1 \cap W_2$.

Since $\mathbf{v} \in W_1 \cap W_2$, it can be written uniquely as a linear combination of the basis B of the intersection:

$$\mathbf{v} = \sum_{i=1}^n \delta_i \mathbf{b}_i$$

Equating this with the RHS expression of $\mathbf{v}$:

$$\sum_{i=1}^n \delta_i \mathbf{b}_i = - \sum_{k=1}^q \gamma_k \mathbf{t}_k \implies \sum_{i=1}^n \delta_i \mathbf{b}_i + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0}.$$

The set $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 and is therefore linearly independent. This forces all coefficients to be zero. Specifically, all $\gamma_k = 0$.

Substitute $\gamma_k = 0$ back into 1.8.2:

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = \mathbf{0}$$

Since $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 , it is linearly independent. This forces all $\alpha_i = 0$ and all $\beta_j = 0$.

Since all α, β, γ are zero, S is linearly independent. ■

Use dimension counting, since S is a basis for $W_1 + W_2$:

$$\begin{aligned}\dim(W_1 + W_2) &= |S| \\ &= n + p + q \\ &= (n + p) + (n + q) - n \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).\end{aligned}$$

■

Remark. **Counter-example for 3 Subspaces**

It is a coincidence that this formula resembles the “Inclusion-Exclusion Principle” from set theory. This intuition fails for three or more subspaces. The formula:

$$\dim(W_1 + W_2 + W_3) \neq \sum \dim(W_i) - \sum \dim(W_i \cap W_j) + \dim(W_1 \cap W_2 \cap W_3)$$

Counter-example: Consider three distinct lines through the origin in $\mathbb{R}^2$ (e.g., x-axis, y-axis, and $y = x$).

- LHS: The sum of three lines fills the plane, so dimension is **2**.
- RHS: $1 + 1 + 1 - (0 + 0 + 0) + 0 = 3$.

Since $2 \neq 3$, the formula is false.

1.8.3 Direct Sum of Two Subspaces

Definition 1.8.2 — Direct Sum of Two Subspaces

Let W_1, W_2 be subspaces of a vector space V . If $W_1 \cap W_2 = \{\mathbf{0}\}$, then we write $W_1 + W_2$ as

$$W_1 \oplus W_2$$

and call it the **direct sum** of W_1 and W_2 .

Example.

Consider the vector space $V = M_n(\mathbb{C})$, and two of its subspaces:

- $W_1 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = \mathbf{A}\}$ (the subspace of symmetric matrices);
- $W_2 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = -\mathbf{A}\}$ (the subspace of skew-symmetric matrices).

We can derive that $V = W_1 \oplus W_2$.

Proof. We first show that $V = W_1 + W_2$. That is, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix. Let $\mathbf{A} \in M_n(\mathbb{C})$, notice that we have the identity:

$$\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$$

where the first term is symmetric and the second term is skew-symmetric. Therefore, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix.

To establish that $W_1 \cap W_2 = \{\mathbf{O}\}$, let $\mathbf{A} \in W_1 \cap W_2$. By definition, this implies $\mathbf{A} \in W_1$ and $\mathbf{A} \in W_2$. Consequently, the identities $\mathbf{A} = \mathbf{A}^\top$ and $\mathbf{A} = -\mathbf{A}^\top$ must hold simultaneously. Substituting the former into the latter yields $\mathbf{A} = -\mathbf{A}$. This implies $2\mathbf{A} = \mathbf{O}$, and thus $\mathbf{A} = \mathbf{O}$.

Chapter 2

Linear Transformations

Vector spaces are only half of the story; the other half is the maps between them that respect addition and scalar multiplication. This chapter studies these *linear transformations*.

2.0.1 Overview

In this part, we will explore:

- **Linear transformations:** the definition and basic properties, examples and non-examples, and the fact that a linear transformation is determined by its values on a basis.
- **Kernel and image:** the two subspaces attached to every linear transformation, with examples and their basic theorems.
- **The rank–nullity theorem:** nullity and rank, and the identity $\text{nullity}(f) + \text{rank}(f) = \dim(V)$ for a finite-dimensional V .

2.1 Linear Transformations

In Chapter 1, we studied vector spaces as static objects—sets equipped with addition and scalar multiplication. Now we turn to the study of **structure-preserving maps** between vector spaces. Just as group theory studies homomorphisms and topology studies continuous functions, linear algebra studies **linear transformations**.

A linear transformation is a function that “respects” the vector space structure: it preserves addition and scalar multiplication. This simple requirement has profound consequences. Linear transformations are the morphisms in the category of vector spaces, and understanding them is central to all of linear algebra.

2.1.1 Definition and Basic Properties

Definition 2.1.1 — Linear Transformation

Let V, W be vector spaces over F . A function $T : V \rightarrow W$ is called a **linear transfor-**

tion (or linear map), if

1. $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$, for all $\mathbf{v}_1, \mathbf{v}_2 \in V$;
2. $T(\lambda \mathbf{v}) = \lambda T(\mathbf{v})$, for all $\lambda \in F$ and $\mathbf{v} \in V$.

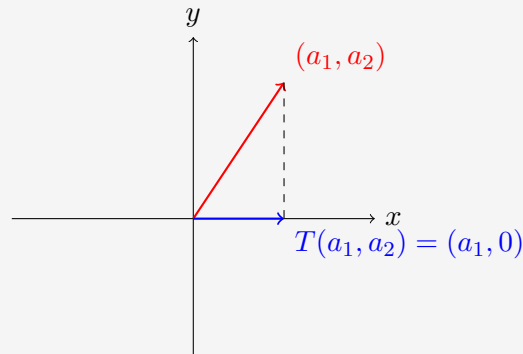
2.1.2 Examples of Linear Transformations

We now present a rich collection of examples. For each, we determine whether the given function is a linear transformation by verifying the two axioms (LT1) and (LT2).

2.1.2.1 Geometric Transformations

Example. (Projection onto a Coordinate Axis)

Determine whether $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, 0)$ is a linear transformation.



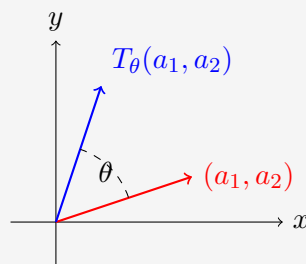
Solution. Let $\mathbf{v} = (a_1, a_2)$ and $\mathbf{w} = (b_1, b_2)$ be vectors in $\mathbb{R}^2$, and let $\lambda \in \mathbb{R}$.

- **Additivity:** $T(\mathbf{v} + \mathbf{w}) = T(a_1 + b_1, a_2 + b_2) = (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0) = T(\mathbf{v}) + T(\mathbf{w})$.
- **Homogeneity:** $T(\lambda \mathbf{v}) = T(\lambda a_1, \lambda a_2) = (\lambda a_1, 0) = \lambda(a_1, 0) = \lambda T(\mathbf{v})$.

Thus T is a linear transformation.

Example. (Rotation in the Plane)

Determine whether rotation by angle θ (counterclockwise) in the plane is a linear transformation. That is, let $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation map.



Solution. Write (a_1, a_2) in polar coordinates: $a_1 = r \cos \phi$ and $a_2 = r \sin \phi$. Then:

$$\begin{aligned} T_\theta(a_1, a_2) &= (r \cos(\theta + \phi), r \sin(\theta + \phi)) \\ &= (r \cos \theta \cos \phi - r \sin \theta \sin \phi, r \sin \theta \cos \phi + r \cos \theta \sin \phi) \\ &= \cos \theta (r \cos \phi, r \sin \phi) + \sin \theta (-r \sin \phi, r \cos \phi) \\ &= \cos \theta (a_1, a_2) + \sin \theta (-a_2, a_1). \end{aligned}$$

Thus $T_\theta(a_1, a_2) = (a_1 \cos \theta - a_2 \sin \theta, a_1 \sin \theta + a_2 \cos \theta)$.

Linearity follows from this explicit formula: for $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$,

$$\begin{aligned} T_\theta(\mathbf{v} + \mathbf{w}) &= \cos \theta (\mathbf{v} + \mathbf{w}) + \sin \theta \cdot J(\mathbf{v} + \mathbf{w}) = T_\theta(\mathbf{v}) + T_\theta(\mathbf{w}), \\ T_\theta(\lambda \mathbf{v}) &= \cos \theta (\lambda \mathbf{v}) + \sin \theta \cdot J(\lambda \mathbf{v}) = \lambda T_\theta(\mathbf{v}), \end{aligned}$$

where $J(a_1, a_2) = (-a_2, a_1)$ is itself linear.

2.1.2.2 Matrix-Induced Transformations

Example. (Left Multiplication by a Matrix)

Let $\mathbf{A} \in M_{m \times n}(F)$ be an $m \times n$ matrix. Determine whether $f_{\mathbf{A}} : F^n \rightarrow F^m$ defined by $f_{\mathbf{A}}(\mathbf{v}) = \mathbf{A}\mathbf{v}$ (where $\mathbf{v}$ is viewed as a column vector) is a linear transformation.

Solution. For $\mathbf{v}, \mathbf{w} \in F^n$ and $\lambda \in F$:

- **Additivity:** $f_{\mathbf{A}}(\mathbf{v} + \mathbf{w}) = \mathbf{A}(\mathbf{v} + \mathbf{w}) = \mathbf{A}\mathbf{v} + \mathbf{A}\mathbf{w} = f_{\mathbf{A}}(\mathbf{v}) + f_{\mathbf{A}}(\mathbf{w})$.
- **Homogeneity:** $f_{\mathbf{A}}(\lambda \mathbf{v}) = \mathbf{A}(\lambda \mathbf{v}) = \lambda(\mathbf{A}\mathbf{v}) = \lambda f_{\mathbf{A}}(\mathbf{v})$.

These follow from the distributive and scalar properties of matrix multiplication.

Example. (A Specific Matrix Transformation)

Determine whether $T : F^3 \rightarrow F^2$ defined by $T(a_1, a_2, a_3) = (2a_1 + a_3, a_2 - a_1)$ is a linear transformation.

Solution. This is the matrix transformation induced by $\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, since

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 2a_1 + a_3 \\ a_2 - a_1 \end{bmatrix}.$$

By the previous example, T is linear.

Example. (Trace)

Determine whether the **trace** function $\text{tr} : M_n(F) \rightarrow F$ defined by $\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii}$ is a linear transformation.

Solution. For matrices $\mathbf{A}, \mathbf{B} \in M_n(F)$ and $\lambda \in F$:

- **Additivity:** $\text{tr}(\mathbf{A} + \mathbf{B}) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$.
- **Homogeneity:** $\text{tr}(\lambda \mathbf{A}) = \sum_{i=1}^n \lambda a_{ii} = \lambda \sum_{i=1}^n a_{ii} = \lambda \text{tr}(\mathbf{A})$.

Thus trace is a linear transformation from the vector space of matrices to the scalar field.

2.1.2.3 Polynomial and Function Space Transformations

Example. (Multiplication by x)

Determine whether $T : F[x] \rightarrow F[x]$ defined by $T(f(x)) = x \cdot f(x)$ is a linear transformation.

Solution. For polynomials $f, g \in F[x]$ and $\lambda \in F$:

- **Additivity:** $T(f + g) = x(f + g) = xf + xg = T(f) + T(g)$.
- **Homogeneity:** $T(\lambda f) = x(\lambda f) = \lambda(xf) = \lambda T(f)$.

Thus multiplication by x is a linear transformation.

Example. (Differentiation)

Determine whether $D : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ defined by $D(f(x)) = f'(x)$ (the derivative) is a linear transformation.

Solution. For polynomials $f, g \in \mathbb{R}[x]$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $D(f + g) = (f + g)' = f' + g' = D(f) + D(g)$.
- **Homogeneity:** $D(\lambda f) = (\lambda f)' = \lambda f' = \lambda D(f)$.

These are the standard sum rule and constant multiple rule from calculus.

Remark. More precisely, $D : \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}[x]_{\leq n-1}$ maps polynomials of degree at most n to polynomials of degree at most $n - 1$.

Example. (Integration)

Determine whether $T : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ defined by $T(f(x)) = \int_0^x f(t) dt$ is a linear transformation.

Solution. For polynomials $f, g \in \mathbb{R}[x]$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $T(f + g) = \int_0^x (f(t) + g(t)) dt = \int_0^x f(t) dt + \int_0^x g(t) dt = T(f) + T(g)$.
- **Homogeneity:** $T(\lambda f) = \int_0^x \lambda f(t) dt = \lambda \int_0^x f(t) dt = \lambda T(f)$.

These follow from the linearity of the integral.

Example. (Shift Operator)

Fix $\theta \in \mathbb{R}$. Determine whether the **shift operator** $S_\theta : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ defined by $S_\theta(f)(x) = f(x - \theta)$ is a linear transformation. (This operator translates the graph of f by θ units to the right.)

Solution. For $f, g \in C(\mathbb{R})$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $S_\theta(f + g)(x) = (f + g)(x - \theta) = f(x - \theta) + g(x - \theta) = S_\theta(f)(x) + S_\theta(g)(x)$.
- **Homogeneity:** $S_\theta(\lambda f)(x) = (\lambda f)(x - \theta) = \lambda f(x - \theta) = \lambda S_\theta(f)(x)$.

Thus the shift operator is linear.

Example. (Composition Operator)

Fix a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$. Determine whether the **composition operator** $C_g : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ defined by $C_g(f) = f \circ g$, i.e., $C_g(f)(x) = f(g(x))$, is a linear transformation.

Solution. For $f, h \in C(\mathbb{R})$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $C_g(f + h)(x) = (f + h)(g(x)) = f(g(x)) + h(g(x)) = C_g(f)(x) + C_g(h)(x)$.
- **Homogeneity:** $C_g(\lambda f)(x) = (\lambda f)(g(x)) = \lambda f(g(x)) = \lambda C_g(f)(x)$.

The shift operator is the special case where $g(x) = x - \theta$.

2.1.2.4 Identity and Inclusion Maps**Example. (Identity Map)**

Let V be a vector space. Determine whether the **identity map** $\text{id}_V : V \rightarrow V$ defined by $\text{id}_V(\mathbf{v}) = \mathbf{v}$ is a linear transformation.

Solution. For $\mathbf{v}, \mathbf{w} \in V$ and $\lambda \in F$:

- **Additivity:** $\text{id}_V(\mathbf{v} + \mathbf{w}) = \mathbf{v} + \mathbf{w} = \text{id}_V(\mathbf{v}) + \text{id}_V(\mathbf{w})$.
- **Homogeneity:** $\text{id}_V(\lambda \mathbf{v}) = \lambda \mathbf{v} = \lambda \text{id}_V(\mathbf{v})$.

The identity map is trivially linear.

Example. (Inclusion Map)

Let $W \subseteq V$ be a subspace. Determine whether the **inclusion map** $\iota : W \hookrightarrow V$ defined by $\iota(\mathbf{w}) = \mathbf{w}$ is a linear transformation.

Solution. The inclusion map is simply the identity map restricted to W , viewed as a map into V . The verification is identical to the identity map.

Example. (Zero Map)

For any vector spaces V and W , determine whether the **zero map** $0 : V \rightarrow W$ defined by $0(\mathbf{v}) = \mathbf{0}_W$ for all $\mathbf{v} \in V$ is a linear transformation.

Solution. For $\mathbf{v}_1, \mathbf{v}_2 \in V$ and $\lambda \in F$:

- **Additivity:** $0(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W = 0(\mathbf{v}_1) + 0(\mathbf{v}_2)$.
- **Homogeneity:** $0(\lambda \mathbf{v}_1) = \mathbf{0}_W = \lambda \mathbf{0}_W = \lambda \cdot 0(\mathbf{v}_1)$.

The zero map is linear.

2.1.2.5 Field-Dependent Examples**Example. (Complex Conjugation)**

Determine whether complex conjugation $T : \mathbb{C} \rightarrow \mathbb{C}$ defined by $T(z) = \bar{z}$, where $\overline{a + bi} = a - bi$, is a linear transformation.

Solution. As a map of $\mathbb{C}$ -vector spaces: T is **not** linear over $\mathbb{C}$.

Consider $z = 1$ and $\alpha = i$. Then:

$$T(\alpha z) = T(i) = -i, \quad \text{but} \quad \alpha T(z) = i \cdot \bar{1} = i.$$

Since $T(i \cdot 1) = -i \neq i = i \cdot T(1)$, the homogeneity condition fails.

As a map of $\mathbb{R}$ -vector spaces: T is linear over $\mathbb{R}$.

For $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$, and $\lambda \in \mathbb{R}$:

- **Additivity:** $T(z + w) = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = (a - bi) + (c - di) = T(z) + T(w)$.
- **Homogeneity:** $T(\lambda z) = \overline{\lambda a + \lambda bi} = \lambda a - \lambda bi = \lambda(a - bi) = \lambda T(z)$.

This example illustrates that **linearity depends on the choice of scalar field**.

2.1.3 Non-Examples of Linear Transformations

It is equally important to recognize functions that are **not** linear transformations. We verify non-linearity by finding a counterexample to either (LT1) or (LT2).

Example. (Squaring)

Determine whether $T : F^2 \rightarrow F^2$ defined by $T(a_1, a_2) = (a_1^2, 0)$ is a linear transformation.

Solution. We show that T fails additivity. Consider $\mathbf{v} = (1, 0)$:

$$T(\mathbf{v}) + T(\mathbf{v}) = (1, 0) + (1, 0) = (2, 0),$$

but

$$T(\mathbf{v} + \mathbf{v}) = T(2, 0) = (4, 0).$$

Since $(2, 0) \neq (4, 0)$, additivity fails and T is not linear.

Alternatively, homogeneity fails: $T(2\mathbf{v}) = T(2, 0) = (4, 0) \neq 2 \cdot (1, 0) = (2, 0) = 2T(\mathbf{v})$.

Example. (Norm)

Determine whether $T : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $T(\mathbf{v}) = \|\mathbf{v}\|$ (the Euclidean norm) is a linear transformation.

Solution. Homogeneity fails for negative scalars. Consider $\mathbf{v} = (1, 0, \dots, 0)$ and $\lambda = -1$:

$$T(\lambda\mathbf{v}) = T(-1, 0, \dots, 0) = \|(-1, 0, \dots, 0)\| = 1,$$

but

$$\lambda T(\mathbf{v}) = (-1) \cdot \|(1, 0, \dots, 0)\| = -1.$$

Since $1 \neq -1$, homogeneity fails and T is not a linear transformation.

(The norm does satisfy $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$, but this is not the same as $c\|\mathbf{v}\|$ when $c < 0$.)

2.1.4 Basic Theorems

Theorem 2.1.1 — Zero Maps to Zero

Let $T : V \rightarrow W$ be a linear transformation. Then $T(\mathbf{0}_V) = \mathbf{0}_W$.

Proof. Using additivity with $\mathbf{v} = \mathbf{0}_V$:

$$T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V).$$

Adding $-T(\mathbf{0}_V)$ to both sides gives $\mathbf{0}_W = T(\mathbf{0}_V)$. ■

Remark. This theorem provides a quick test for non-linearity: if $T(\mathbf{0}) \neq \mathbf{0}$, then T is not linear.

Example. (Constant Function)

Determine whether $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = 5$ for all $x \in \mathbb{R}$ is a linear transformation.

Solution. We use 2.1.1: any linear transformation must map zero to zero.

Here, $T(0) = 5 \neq 0$.

Therefore, T is not a linear transformation.

Example. (Translation)

Determine whether $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{v}) = \mathbf{v} + \mathbf{c}$ for some fixed nonzero $\mathbf{c} \in \mathbb{R}^2$ is a linear transformation.

Solution. Using 2.1.1: $T(\mathbf{0}) = \mathbf{0} + \mathbf{c} = \mathbf{c} \neq \mathbf{0}$.

Therefore, translation by a nonzero vector is not linear.

Maps of the form $\mathbf{v} \mapsto \mathbf{A}\mathbf{v} + \mathbf{c}$ are called **affine transformations**. They are linear if and only if $\mathbf{c} = \mathbf{0}$.

Theorem 2.1.2 — Linear Maps Preserve Negation

Let $T : V \rightarrow W$ be a linear transformation. Then $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v} \in V$.

Proof. Using homogeneity with $\lambda = -1$:

$$T(-\mathbf{v}) = T((-1)\mathbf{v}) = (-1)T(\mathbf{v}) = -T(\mathbf{v}).$$

■

Theorem 2.1.3 — Equivalent Characterization of Linearity

Let V, W be vector spaces over F and $T : V \rightarrow W$ be a function. Then T is a linear transformation if and only if

$$T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2)$$

for all $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$.

Proof. ($\Rightarrow$) Suppose T is linear. Then for any $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$:

$$T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = T(\alpha\mathbf{v}_1) + T(\beta\mathbf{v}_2) \quad (\text{by LT1})$$

$$= \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2) \quad (\text{by LT2})$$

($\Leftarrow$) Suppose $T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2)$ for all $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$.

- **Additivity (LT1):** Setting $\alpha = \beta = 1$, we get $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$.
- **Homogeneity (LT2):** Setting $\beta = 0$, we get $T(\alpha\mathbf{v}_1 + \mathbf{0}) = \alpha T(\mathbf{v}_1) + 0 \cdot T(\mathbf{v}_2)$, i.e., $T(\alpha\mathbf{v}_1) = \alpha T(\mathbf{v}_1)$.

Thus T satisfies both (LT1) and (LT2), so T is linear.

■

Theorem 2.1.4 — Linear Maps Preserve Linear Combinations

Let $T : V \rightarrow W$ be a linear transformation. Then for any vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ and scalars $a_1, \dots, a_n \in F$:

$$T\left(\sum_{i=1}^n a_i \mathbf{v}_i\right) = \sum_{i=1}^n a_i T(\mathbf{v}_i).$$

Proof. By induction on n . The base case $n = 1$ is homogeneity. For the inductive step:

$$\begin{aligned} T\left(\sum_{i=1}^{n+1} a_i \mathbf{v}_i\right) &= T\left(\sum_{i=1}^n a_i \mathbf{v}_i + a_{n+1} \mathbf{v}_{n+1}\right) \\ &= T\left(\sum_{i=1}^n a_i \mathbf{v}_i\right) + T(a_{n+1} \mathbf{v}_{n+1}) && \text{(additivity)} \\ &= \sum_{i=1}^n a_i T(\mathbf{v}_i) + a_{n+1} T(\mathbf{v}_{n+1}) && \text{(induction + homogeneity)} \\ &= \sum_{i=1}^{n+1} a_i T(\mathbf{v}_i). \end{aligned}$$

■

2.1.5 Linear Transformation Determined by Basis

Another useful fact of a linear transformation is that once we know how it acts on a basis, then we know how it acts on every vector in the domain.

Theorem 2.1.5 — Linear Transformations are Determined by Their Action on a Basis

Let $T : V \rightarrow W$ be a linear transformation and V is finite dimensional, and $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then we have:

1. For every $\mathbf{v} \in V$, $f(\mathbf{v})$ is a linear combination of $f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)$.
2. Suppose the linear transformation $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$ for all i . Then $g = f$.

We say ‘ f is determined by $f(\mathbf{v}_1), \dots, f(\mathbf{v}_n)$ ’ in this case.

Proof. Let $\mathbf{v} \in V$. Since B is a basis of V , we can write $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$. Then

$$\begin{aligned} f(\mathbf{v}) &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n). \end{aligned}$$

This proves the first condition.

Moreover, if $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$, then:

$$\begin{aligned} g(\mathbf{v}) &= g(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n) \\ &= \alpha_1 g(\mathbf{v}_1) + \cdots + \alpha_n g(\mathbf{v}_n) \\ &= \alpha_1 f(\mathbf{v}_1) + \cdots + \alpha_n f(\mathbf{v}_n) \\ &= f(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n) \\ &= f(\mathbf{v}). \end{aligned}$$

Since $g(\mathbf{v}) = f(\mathbf{v})$ for all $\mathbf{v} \in V$, we have $f = g$. This finishes the second condition. ■

Example.

Is it possible to have a linear map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that

$$f(1, 0) = (2, 3, 4), \quad f(1, 1) = (3, 4, 5), \quad f(2, 3) = (4, 56)?$$

Solution. Idea: Since $\{(1, 0), (1, 1)\}$ forms a basis of $\mathbb{R}^2$, these two function value would determine f completely.

Suppose such a linear transformation f exists. Observe that

$$(2, 3) = -1(1, 0) + 3(1, 1).$$

Apply f then we have:

$$\begin{aligned} f(2, 3) &= f(-1(1, 0) + 3(1, 1)) \\ &= -1f(1, 0) + 3f(1, 1) \\ &= -1(2, 3, 4) + 3(3, 4, 5) \\ &= (7, 8, 11) \\ &\neq (4, 56). \end{aligned}$$

Which is inconsistent with the third equation, therefore no such transformation exists.

Example.

Suppose we are give a linear transformation $f : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$. And it is also know that

$$f(x^3) = 3x^2, \quad f(x^2) = 2x, \quad f(x) = 1, \quad f(1) = 0.$$

Solution. As $\{1, x, x^2, x^3\}$ forms a basis of $\mathbb{R}[x]_{\leq 3}$. And $D(x^k) = f(x^k)$, for $k = 0, 1, 2, 3$, as D is differentiation. We have that $D \equiv f$ is the same linear transformation.

2.2 Kernel and Image

2.2.1 Definition

Definition 2.2.1 — Kernel

Let $f : V \rightarrow W$ be a linear transformation. Define the **kernel** of f by

$$\ker(f) := \{\mathbf{v} \in V : f(\mathbf{v}) = \mathbf{0}_W\}.$$

i.e. The kernel contains everything in the domain that becomes the zero vector under f .

Definition 2.2.2 — Image

Let $f : V \rightarrow W$ be a linear transformation. Define the **image** of f by

$$\text{im}(f) := \{\mathbf{w} \in W : \mathbf{w} = f(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}.$$

i.e. The image contains all the possible outputs of the function f .

Remark. The kernel is also known as the nullspace, which you might be familiar from Gilbert Strang's book. And the image is also called the range. Also, we sometimes denote the image as $f(V)$.

2.2.2 Examples

Example. (Differentiation)

Let $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ be defined by $D(f(x)) = f'(x)$. Find the kernel and image of D .

Solution. Kernel: We seek all polynomials $f(x) \in \mathbb{R}[x]_{\leq 3}$ such that $D(f(x)) = f'(x) = 0$.

A polynomial has derivative zero if and only if it is a constant. Thus:

$$\ker(D) = \{c : c \in \mathbb{R}\} = \text{span}\{1\}.$$

So $\dim(\ker(D)) = 1$.

Image: We need to determine which polynomials in $\mathbb{R}[x]_{\leq 2}$ can be obtained as derivatives.

Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \in \mathbb{R}[x]_{\leq 3}$. Then:

$$D(f(x)) = f'(x) = a_1 + 2a_2x + 3a_3x^2.$$

As a_1, a_2, a_3 range over all real numbers, we can obtain any polynomial of degree at most 2. Therefore:

$$\text{im}(D) = \mathbb{R}[x]_{\leq 2}.$$

So $\dim(\text{im}(D)) = 3$.

Note that $\dim(\ker(D)) + \dim(\text{im}(D)) = 1 + 3 = 4 = \dim(\mathbb{R}[x]_{\leq 3})$.

Example. (Trace)

Let $\text{tr} : M_2(\mathbb{R}) \rightarrow \mathbb{R}$ be defined by $\text{tr}(\mathbf{A}) = a_{11} + a_{22}$. Find the kernel and image of tr .

Solution. Kernel: We seek all matrices $\mathbf{A} \in M_2(\mathbb{R})$ such that $\text{tr}(\mathbf{A}) = 0$.

Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\text{tr}(\mathbf{A}) = a + d = 0$, which means $d = -a$.

So the kernel consists of all matrices of the form:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Therefore:

$$\ker(\text{tr}) = \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}.$$

So $\dim(\ker(\text{tr})) = 3$.

Image: For any $r \in \mathbb{R}$, we can find a matrix with trace r , for example $\begin{bmatrix} r & 0 \\ 0 & 0 \end{bmatrix}$.

Therefore:

$$\text{im}(\text{tr}) = \mathbb{R}.$$

So $\dim(\text{im}(\text{tr})) = 1$.

Note that $\dim(\ker(\text{tr})) + \dim(\text{im}(\text{tr})) = 3 + 1 = 4 = \dim(M_2(\mathbb{R}))$.

2.2.3 Basic Theorems

Theorem 2.2.1 — Properties of Kernel

Let $f : V \rightarrow W$ be a linear transformation. Then:

1. $\ker(f)$ is a subspace of V ;
2. f is injective iff $\ker(f) = \{\mathbf{0}_V\}$.

Proof.

1.
 - Non-emptiness: As f is linear, $f(\mathbf{0}_V) = \mathbf{0}_W$, therefore $\mathbf{0}_V \in \ker(f)$.
 - Closure under addition: Let $\mathbf{v}_1, \mathbf{v}_2 \in \ker(f)$, it implies that $f(\mathbf{v}_1) = \mathbf{0}_W$ and

$f(\mathbf{v}_2) = \mathbf{0}_W$. We have

$$f(\mathbf{v}_1 + \mathbf{v}_2) = f(\mathbf{v}_1) + f(\mathbf{v}_2) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W.$$

Therefore $\mathbf{v}_1 + \mathbf{v}_2 \in \ker(f)$.

- Closure under scalar multiplication: Let $\mathbf{v} \in \ker(f)$, it implies that $f(\mathbf{v}) = \mathbf{0}_W$. We have

$$f(\lambda \mathbf{v}) = \lambda f(\mathbf{v}) = \lambda \mathbf{0}_W = \mathbf{0}_W.$$

Therefore $\lambda \mathbf{v} \in \ker(f)$.

2. • ($\Rightarrow$) Suppose f is injective. Let $\mathbf{v} \in \ker(f)$, we have

$$\begin{aligned} f(\mathbf{v}) &= \mathbf{0}_W \\ &= f(\mathbf{0}_V) \end{aligned} \quad (\text{linearity})$$

By having $f(\mathbf{v}) = f(\mathbf{0}_V)$ and f is injective, it implies that $\mathbf{v} = \mathbf{0}_V$, so $\ker(f) = \{\mathbf{0}_V\}$.

- ($\Leftarrow$) Suppose $\ker(f) = \{\mathbf{0}_V\}$, and we consider $f(\mathbf{v}) = f(\mathbf{w})$. Reordering the terms we get $f(\mathbf{v}) - f(\mathbf{w}) = \mathbf{0}_W$, and since f is linear we have $f(\mathbf{v} - \mathbf{w}) = \mathbf{0}_W$. Therefore $\mathbf{v} - \mathbf{w} \in \ker(f) = \{\mathbf{0}_V\}$, implying that $\mathbf{v} - \mathbf{w} = \mathbf{0}_V$, hence $\mathbf{v} = \mathbf{w}$. So f is injective.

■

Theorem 2.2.2 — Properties of Image

Let $f : V \rightarrow W$ be a linear transformation. Then:

1. $\text{im}(f)$ is a subspace of W ;
2. f is surjective iff $\text{im}(f) = W$.

Proof.

1. • Non-emptiness: Since f is linear, $\mathbf{0}_W = f(\mathbf{0}_V)$, therefore $\mathbf{0}_W \in \text{im}(f)$.
• Closure under addition: Let $\mathbf{w}_1, \mathbf{w}_2 \in \text{im}(f)$. So $\mathbf{w}_1 = f(\mathbf{v}_1)$, and $\mathbf{w}_2 = f(\mathbf{v}_2)$ for some $\mathbf{v}_1, \mathbf{v}_2 \in V$. Notice that

$$\mathbf{w}_1 + \mathbf{w}_2 = f(\mathbf{v}_1) + f(\mathbf{v}_2) = f(\mathbf{v}_1 + \mathbf{v}_2).$$

Therefore $\mathbf{w}_1 + \mathbf{w}_2 \in \text{im}(f)$.

- Closure under scalar multiplication: Let $\mathbf{w} \in \text{im}(f)$. So $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$. Notice that

$$\lambda \mathbf{w} = \lambda f(\mathbf{v}) = f(\lambda \mathbf{v}).$$

Therefore $\lambda \mathbf{w} \in \text{im}(f)$.

2. f is surjective $\Leftrightarrow \forall \mathbf{w} \in W$, we have $\mathbf{w} \in \text{im}(f)$ for some $\mathbf{v} \in V$, that is equivalent to say $W \subseteq \text{im}(f)$, and by the first condition we know that $\text{im}(f) \subseteq W$ is always true. Hence we have f is surjective $\Leftrightarrow W = \text{im}(f)$.

■

2.3 Rank-Nullity Theorem

2.3.1 Nullity and Rank

Definition 2.3.1 — Nullity

Let $f : V \rightarrow W$ be a linear transformation. The **nullity** of f is defined by

$$\text{nullity}(f) := \dim(\ker(f)).$$

Definition 2.3.2 — Rank

Let $f : V \rightarrow W$ be a linear transformation. The **rank** of f is defined by

$$\text{rank}(f) := \dim(\text{im}(f)).$$

Example.

Recall that for $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x),$$

we have $\ker(D) = \mathbb{R}$ and $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$. Compute the nullity and image of D .

Solution. $\text{nullity}(D) := \dim_{\mathbb{R}}(\ker(D)) = \dim_{\mathbb{R}}(\mathbb{R}) = 1.$

$\text{rank}(D) := \dim_{\mathbb{R}}(\text{im}(D)) = \dim_{\mathbb{R}}(\mathbb{R}[x]_{\leq 2}) = 3.$

Remark. For other examples of nullity and rank, review the examples in the last section, which we have it covered.

2.3.2 The Theorem

Theorem 2.3.1 — Rank-Nullity Theorem

Let V be a finite dimensional vector space and $f : V \rightarrow W$ be a linear transformation. Then we have

$$\text{nullity}(f) + \text{rank}(f) = \dim(V).$$

Proof. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$, having $\text{nullity}(f) = \dim(\ker f) = n$. Since $\ker(f) \subseteq V$, by Basis Extension Theorem, we can extend S to $S' = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\}$, a basis of V , having $\dim(V) = n + k$.

Claim. $S'' = \{f(\mathbf{s}_1), f(\mathbf{s}_2), \dots, f(\mathbf{s}_k)\}$ is a basis of $\text{im}(f)$.

Proof. We verify two properties.

Spanning. Let $\mathbf{w} \in \text{im}(f)$, we have $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$ by definition. As S' is a basis of V , we can write $\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i + \sum_{i=1}^k \beta_i \mathbf{s}_i$. Then we apply f to both sides:

$$\mathbf{w} = f(\mathbf{v}) = f\left(\sum \alpha_i \mathbf{v}_i + \sum \beta_i \mathbf{s}_i\right) = \sum \alpha_i f(\mathbf{v}_i) + \sum \beta_i f(\mathbf{s}_i) = \sum \beta_i f(\mathbf{s}_i),$$

where the last equality holds since $\mathbf{v}_i \in \ker(f)$. Since $\mathbf{w}$ can be expressed by $f(\mathbf{s}_i)$'s, therefore S'' spans $\text{im}(f)$.

Linear independence. Suppose that $\alpha_1 f(\mathbf{s}_1) + \cdots + \lambda_k f(\mathbf{s}_k) = \mathbf{0}$. By linearity of f , it becomes $f(\alpha_1 \mathbf{s}_1 + \cdots + \alpha_k \mathbf{s}_k) = \mathbf{0}$. Therefore we have $\alpha_1 \mathbf{s}_1 + \cdots + \alpha_k \mathbf{s}_k \in \ker(f)$. Moreover, as S is a basis of $\ker(f)$, we know that $\sum \alpha_i \mathbf{s}_i = \sum \beta_i \mathbf{v}_i$ for some $\beta_i \in F$. Reordering gives $\sum \alpha_i \mathbf{s}_i - \sum \beta_i \mathbf{v}_i = \mathbf{0}$. As S' is linearly independent, this forces all α_i 's and β_i 's to be 0. This completes the proof of the claim. ■

Hence, we have

$$\text{rank}(f) = \dim(\text{im}(f)) = k = (n + k) - n = \dim(V) - \text{nullity}(f).$$
■